

Programming with LLMs

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Abstract

The rise in mental health concerns has highlighted the need for accessible and scalable solutions that provide emotional support and guidance. MindMate is an innovative mental health chatbot designed to function as both a therapeutic companion and a supportive friend for individuals in need. Utilizing advanced natural language processing (NLP) techniques, MindMate has been trained on multiple datasets curated for empathy, emotional intelligence, and conversational understanding. The chatbot engages users in meaningful, empathetic conversations, offering therapeutic insights and coping strategies while maintaining a compassionate tone.

A unique feature of MindMate is its integration with a live call option, allowing users to seamlessly transition from chatting with the bot to speaking with a highly qualified therapist or mental health professional, thereby combining automated and human support. This hybrid approach addresses the limitations of chatbot-only interactions by ensuring that users have access to professional assistance when required. MindMate aims to bridge the gap between traditional therapy and technology-driven solutions, providing a scalable, user-friendly platform for individuals seeking mental health support.

This paper delves into the design, development, and evaluation of MindMate, covering the chatbot’s underlying architecture, datasets used for training, ethical considerations, and the integration of real-time human assistance through teleconferencing. The project also compares the performance of multiple large language models (LLMs), selecting the model that yields the best results based on response quality, emotional understanding, and user feedback. While broad user testing will not be conducted, the survey conducted to collect user feedback will provide insights into the chatbot’s efficacy and potential as a supplementary tool for mental health care.

1 Introduction

Mental health has become a growing global concern, with millions of individuals experiencing conditions such as anxiety, depression, stress, and loneliness. Despite this widespread need, access to professional mental health services remains a significant challenge. Traditional therapy is often expensive, stigmatized, and in many areas, there is a severe shortage of qualified mental health professionals. These barriers create a gap between those seeking help and the support they require.

Recognizing this critical issue, we set out to develop MindMate, a mental health chatbot designed to provide accessible, on-demand emotional support and therapeutic guidance.

The project was driven by a desire to address the limitations of existing mental health care systems and offer an innovative solution that could potentially scale to meet the needs of a diverse range of users. With rapid advancements in artificial intelligence (AI) and natural language processing (NLP), the opportunity to leverage technology in mental health care has never been more feasible or timely.

MindMate is designed to function similarly to a therapist or a supportive friend, bridging the gap between human empathy and machine intelligence. Unlike many chatbots that operate on predefined scripts, MindMate is trained on multiple datasets specifically tailored for emotional understanding, allowing it to provide responses that are empathetic, sensitive, and aligned with therapeutic practices. The chatbot engages users in open-ended conversations, offering coping mechanisms, emotional validation, and mental health resources while creating a sense of companionship that is crucial in moments of distress.

What sets MindMate apart is its hybrid approach. Recognizing that no machine can fully replace the nuance and expertise of a trained therapist, we integrated a call feature that allows users to connect with an actual therapist or highly qualified mental health professional. This option is crucial for users who may need immediate human intervention or prefer speaking directly with someone who can offer specialized support. This feature not only adds a layer of safety but also creates a more holistic experience for the user, allowing them to transition seamlessly between the chatbot’s automated support and professional human interaction.

One of the key objectives of this project is to determine which large language model (LLM) is most effective for managing mental health conversations. We explored several models, including GPT-4oOpenAI (2023), Meta’s LLaMA 2Touvron et al. (2023a), LLaMA 3Touvron et al. (2023b), and GeminiDeepMind (2023), comparing their performance based on metrics such as response quality, emotional depth, and the model’s ability to maintain compassionate and meaningful conversations. These models were selected for their state-of-the-art capabilities in natural language understanding and generation, but their effectiveness in the context of mental health required a more nuanced evaluation.

- GPT-3.5 and GPT-4o, developed by OpenAI, are known for their conversational capabilities, with GPT-4o offering greater accuracy and deeper understanding in complex conversations.
- Meta’s LLaMA 2 and LLaMA 3 offer open-access models, designed with a focus on scalability and improved memory, making them ideal candidates for large-scale deployment.
- Gemini, developed by Google DeepMind, is a state-of-the-art model designed to excel in generating high-quality text and understanding context across various applications. It offers advanced capabilities in language understanding and generation, making it a strong contender in the field of large language models.

Through a comparative analysis, we aim to select the model that delivers the best results in terms of emotional intelligence, contextual understanding, and overall user satisfaction. This LLM will be used to handle general and mental health-related conversations, ensuring

that MindMate is both functional and empathetic in its responses.

The need for a platform like MindMate stems from both societal and technological trends. With the increasing prevalence of mental health issues, there is a clear demand for accessible mental health resources that can provide real-time assistance. Additionally, AI-driven chatbots have already been successfully implemented in customer service, education, and entertainment industries, proving their versatility and ability to simulate human-like interactions. MindMate takes these innovations and applies them to the sensitive domain of mental health care, aiming to provide users with a tool that is always available, non-judgmental, and trained to listen.

This paper explores the development and functionality of MindMate, detailing the steps taken to design its architecture, the datasets used for training, and the ethical considerations surrounding AI-driven mental health solutions. Additionally, the incorporation of the call feature ensures that MindMate not only acts as a supplement to traditional therapy but also offers users the opportunity to access real-time professional help when needed.

Through this project, we hope to contribute to the ongoing conversation about the role of technology in mental health care and demonstrate how a well-designed chatbot can provide meaningful support, whether it is for someone experiencing mild stress or someone in need of more intensive care.

2 Datasets

In developing MindMate, our primary focus was ensuring that the chatbot could effectively handle conversations related to mental health. The training process was guided by a need to create a system capable of understanding and responding empathetically to users who may be experiencing a wide range of emotional states, from mild stress to severe anxiety and depression. To achieve this, we exclusively trained MindMate on datasets comprising mental health-related dialogues, intentionally avoiding datasets that dealt with casual or general conversations. This selective training ensured that the chatbot remained highly specialized in the domain of mental health, while also posing certain early limitations.

2.1 Initial Training and Challenges

At the outset, the chatbot’s training was confined to datasets consisting only of mental health-related conversations. This focus meant that while MindMate excelled at recognizing and responding to complex emotional cues, it initially struggled with more commonplace interactions, such as greeting a user or answering general questions unrelated to mental health. A user might begin a conversation with a casual "Hello" or "How are you?", and the chatbot lacking exposure to such exchanges would either not respond appropriately or attempt to steer the conversation back to mental health topics.

To address this, we adopted a hybrid model by integrating GPT to handle these non-mental health interactions. GPT, being a more general-purpose conversational model, was tasked with managing the introductory, small-talk segments of a conversation, ensuring that the user experience remained fluid and engaging. Meanwhile, when the conversation transitioned into mental health concerns, MindMate’s specialized training took over. This

solution allowed us to retain the chatbot’s expertise in mental health while ensuring it could handle the day-to-day nuances of conversation.

2.2 Datasets Used

For the core mental health training, we employed four distinct datasets, each chosen for its focus on psychological well-being, therapeutic conversations, and empathetic responses. Below are the datasets used in training MindMate:

- **Psych8k Counseling Recordings Dataset:Liu et al. (2023)**

The data used for this project comes from approximately 260 real conversations in counseling sessions. The transcripts of these conversations, covering topics such as emotion, family, relationships, career development, and academic stress, were used as the primary source for building the training and testing datasets. This dataset contains 8,187 rows and provides MindMate with the ability to understand and respond to complex, in-depth discussions typical of professional counseling environments.

This dataset, known as Psych8k, was also used in a research paper that presented a model named ChatCounselor. The paper emphasized that unlike generic chatbots, ChatCounselor’s foundation in real counseling sessions provided it with specialized knowledge and counseling skills, making it a highly domain-specific model for mental health support. Constructed from 260 in-depth, one-hour counseling interviews, Psych8k formed the backbone of the dataset used in our project, ensuring high-quality, professional-level responses from the chatbot.

Evaluations conducted using a set of real-world counseling questions showed that models trained on Psych8k, including ChatCounselor, outperformed open-source alternatives and came close to ChatGPT in terms of counseling response quality. This dataset was crucial for training MindMate to handle the nuances of mental health discussions with accuracy and empathy.

- **Counsel-Chat: CounselChat Forum Scrape:Bertagnolli (2020)**

This dataset was scraped from CounselChat.com, a platform where licensed clinicians respond to mental health-related questions posed by users. The dataset includes 2,775 rows of expert responses, which helped train MindMate to offer advice similar to that provided by professionals in real-world therapeutic settings.

The origins of this dataset stem from a broader effort to improve the quality of mental health care through Natural Language Processing (NLP). One of the challenges in this domain is the scarcity of high-quality data that mirrors real therapist-client interactions. Many NLP projects have resorted to scraping social platforms like Reddit, but these interactions are often considered poor substitutes for actual therapy sessions.

This dataset was designed to provide a solution by offering authentic therapist responses to real mental health questions, sourced directly from CounselChat. The collaboration with Dr. Grin Lord and CounselChat co-founders, Eric Ström and Phil Lee, resulted in a unique resource for developing mental health chatbots. Its inclusion in MindMate’s training allowed the model to provide expert-level therapeutic advice.

- **Mental health conversational dataset: Mental Health Q&A from Healthcare Blogs:Anonymous (2024)**

This dataset consists of 175 rows of conversational question-and-answer pairs, where patients ask mental health-related questions, and healthcare providers offer their responses. The data was curated from leading healthcare websites such as WebMD, Mayo Clinic, and HealthLine, making it an invaluable resource for building MindMate’s understanding of common mental health concerns. All the conversations in this dataset were anonymized to ensure no personally identifiable information (PII) was present.

The rationale behind curating this dataset was to supplement conversational AI models with accurate and compassionate responses to mental health questions. Chatbots provide immediate, empathetic support, which can be crucial in moments of distress. While not a replacement for human interaction, the model trained on this dataset can act as a helpful supplement to those seeking mental health assistance. The dataset’s structure—a simple text format containing paired questions and answers—made it ideal for fine-tuning MindMate’s conversational skills in a healthcare context.

- **Mental health counseling conversations: Online Counseling and Therapy Platforms dataset:onl (2024)**

This dataset contains 3,512 rows of questions and answers from two different online therapy and counseling platforms. The questions encompass a broad range of mental health topics, and the answers are provided by qualified psychologists, giving the dataset a professional and expert-level depth.

The dataset is structured to support text generation tasks, particularly in generating advice or suggestions in response to mental health-related questions. Each data instance includes two fields: Context (the user’s question) and Response (the psychologist’s answer). This clear, structured format made it straightforward to incorporate into MindMate’s training. By using this dataset, we aimed to further enhance the chatbot’s ability to give well-rounded, professional advice.

The dataset creation focused on ensuring the data was cleaned and anonymized to remove any personal or sensitive information, following ethical guidelines for handling mental health data. It provided a rich source for developing a model capable of delivering psychological advice with a high degree of accuracy and empathy.

2.3 Dataset Cleaning Procedure

A crucial step in preparing the data for training MindMate was ensuring that the datasets were properly structured and cleaned to enhance the quality of the model’s responses. Given that the conversations within the datasets came from real counseling sessions, forums, and mental health platforms, it was essential to clearly delineate the roles of user/patient and therapist in each interaction. The cleaning process involved:

1. **Dividing Conversations:** We carefully separated the text into distinct question-and-answer pairs. In each conversation, the user/patient’s input was identified as the question, while the therapist’s response was labeled as the answer. This ensured that

the model could learn how to effectively respond to mental health-related queries, distinguishing between the two roles.

2. **Formatting for Clarity:** After division, the data was restructured into a consistent format to avoid any overlap or confusion between roles. Each instance was labeled to ensure that the model would clearly differentiate between the question (coming from the user) and the answer (generated by the model to emulate the therapist).
3. **Removing Unnecessary Information:** During the cleaning process, any additional information that was irrelevant to the training process was removed. This step ensured that only relevant therapeutic exchanges were retained, optimizing the data for more effective training.

This systematic cleaning process helped streamline the training phase, enabling the models to better understand and respond within the framework of a therapeutic conversation. By structuring the data in this manner, MindMate was better equipped to generate meaningful, empathetic, and relevant responses based on the user’s input.

2.4 Refining Conversational Balance

By combining these datasets, we created a chatbot that is well-versed in various therapeutic approaches while remaining adaptable to the conversational style of a wide range of users. The integration of GPT into the system allowed for smooth transitions between mental health topics and general conversation, improving user experience without sacrificing the chatbot’s core function as a mental health assistant.

Through this targeted training process, we achieved a balance where MindMate could function not only as a specialized tool for mental health but also as a companion capable of engaging in more general interactions when necessary.

3 System Overview

MindMate is a mental health chatbot designed to assist users by providing empathetic conversations and offering the option to speak to a live therapist. The system consists of multiple components that work together to deliver a seamless user experience, including a chatbot powered by large language models (LLMs), a UI designed with mental health considerations, and a call feature that connects users to therapists. The following sections provide a detailed overview of the system’s architecture, components, and key functionalities.

3.1 System Architecture

MindMate operates as a web-based application where the frontend and backend work in synergy to provide a seamless user experience. The frontend, built using FlaskRonacher (2023) and HTML, serves as the user’s primary interface, offering a real-time chat interface and a “Call” button that allows for live video sessions. This interface has been designed with user mental well-being in mind, featuring bright, soothing visuals and optional background music that can be toggled on or off.

On the backend, the system processes incoming messages, leverages large language models (LLMs) for generating responses, and integrates third-party services like JitsiTeam (2023a) for video calls. The LLMs are responsible for producing context-aware, empathetic responses, with general queries handled by GPT models and mental health-specific queries by custom-trained datasets. Additionally, the backend manages room assignments for the call feature, ensuring a smooth transition from chatbot interaction to human assistance.

The architecture consists of two main components:

- **User Interaction (Frontend):** A user-friendly interface that supports conversational exchange with the chatbot, combined with a well-timed “Call” feature for initiating live video consultations.
- **Backend Services:** Responsible for natural language understanding, response generation using LLMs, and handling all aspects of video call sessions through the Jitsi API.

Together, these layers ensure that users not only receive real-time chatbot support but also have the option to escalate to a live video consultation when needed

3.2 Components and Modules

1. **Chatbot:** The chatbot uses two language models for conversation generation:
 - **Custom-trained Model:** Focused specifically on mental health-related topics, providing empathetic and therapeutic responses.
 - **GPT (3.5 or 4o):** Handles non-mental health queries and general conversations to ensure versatility in response generation. GPT is invoked when the user’s query is not mental health-related, ensuring the bot remains relevant and helpful across diverse user inputs.
2. **Call Feature:** A core component of MindMate, the call feature allows users to connect with a live therapist or highly qualified moderator after prolonged conversation or upon the user’s request. The call feature is integrated with the Jitsi API, which facilitates video conferencing.
 - **Room Assignment:** Users are assigned to different rooms in a round-robin manner (Rooms 1 to 5), ensuring even distribution across available moderators.
 - **Moderator Notification:** When a user initiates a call, an email is sent to the respective moderator of that room, prompting them to join the call.
3. **Cooldown Mechanism:** After the “Call” button is clicked, a 15-second cooldown period is enforced to prevent users from repeatedly triggering the call feature. During this time, a message appears on the interface stating, “Wait a few seconds while the moderator joins the call.”
4. **UI Elements:** The system’s UI is intentionally designed with mental health in mind, offering a calming experience through the use of bright, soft colors and optional back-

ground music that defaults to 'on'. Users are also informed of the chatbot's limitations, including a warning that it should not replace professional therapy in serious situations.

3.3 Data Flow

The system follows a well-defined flow for handling user interactions:

1. **User Interaction:** The user inputs text via the chat interface.
2. **Processing and Response Generation:** The backend processes the user's input, determining whether it falls within the domain of mental health. If the input is mental health-related, the custom-trained model is used to generate an empathetic response. If the input is general or unrelated to mental health, GPT is invoked.
3. **Call Initiation:** If the user opts for a call, the backend assigns them to one of five rooms, notifies the corresponding moderator, and triggers a 15-second cooldown.
4. **Video Call:** The system generates a Jitsi meeting link for the selected room and sends the link to both the user and moderator. Once both join, the session begins.

3.4 Key Features

- **Empathetic Conversations:** MindMate is designed to offer empathetic, understanding responses. By utilizing custom-trained models on mental health-related datasets, the chatbot can provide more relevant and sensitive responses to users in need of emotional support.
- **Call Feature:** This feature allows users to escalate their conversation from text to video call when they require real-time support from a human moderator. Room assignments and moderator notifications ensure that users are always connected to the appropriate resource.
- **Cooldown Functionality:** The cooldown mechanism prevents spam by introducing a 15-second delay after the call button is clicked, allowing enough time for the moderator to join the video session. A message, "Wait a few seconds while the moderator joins the call," appears to reassure users that the system is working on their request.

3.5 System Workflow

The user journey in MindMate can be summarized as follows:

1. **Initial Conversation:** The user initiates a conversation by interacting with the chatbot through the web-based chat interface. The chatbot processes the input and provides empathetic responses, depending on the nature of the conversation.
2. **Call Activation:** After a certain amount of time or if the conversation becomes more serious, the user has the option to escalate to a live video call by clicking the "Call" button.

3. **Room Assignment and Notification:** The backend assigns the user to a meeting room and sends an email notification to the appropriate moderator.
4. **Video Call:** The user and moderator connect via the Jitsi link provided. The conversation moves from the text-based interface to a live video session.

4 Local Environment: Ollama

OllamaTeam (2023c) is a specialized tool designed to simplify the deployment and execution of large language models (LLMs) like LLaMA on local machines. In our project, we leveraged Ollama for running LLaMA 2 and LLaMA 3 models, allowing us to perform model training, fine-tuning, and inference locally without the need for external cloud services. This provided several advantages in terms of data privacy, computational control, and cost-efficiency.

1. **Advantages of Ollama for Local LLM Execution** Ollama offers a streamlined environment for executing LLaMA models, particularly focused on reducing the typical barriers associated with running large models locally. The following are key benefits:
 - **Ease of Use:** Ollama simplifies the process of running sophisticated models like LLaMA by minimizing the configuration overhead. Its intuitive interface and accessible documentation allowed us to quickly integrate and execute models.
 - **Local Execution:** Running LLaMA 2 and 3 locally provided us with full control over the computational resources, enabling more efficient fine-tuning and experimentation without the latencies or cost overheads associated with cloud-based solutions.
 - **Data Privacy:** By running the models on a local environment, we ensured that sensitive data, particularly the mental health datasets used for training, remained secure and private. This was a critical aspect for a mental health-focused project like MindMate, where user confidentiality is a top priority.
 - **Flexibility in Model Fine-Tuning:** Ollama allowed us to make iterative adjustments to the LLaMA 2 and 3 models in a flexible manner, improving their conversational abilities for mental health-specific interactions.
2. **Performance and Resource Optimization** One of the major benefits of Ollama is its optimization for hardware efficiency. While LLaMA 2 and 3 are relatively large models, Ollama’s design minimizes the resource footprint necessary to run them locally. This made it feasible for us to experiment with different model architectures and configurations without the need for extensive computational resources.
 - **Efficiency on Consumer Hardware:** With Ollama, even high-performance models like LLaMA could run efficiently on consumer-grade hardware, ensuring that the project remained scalable and budget-friendly.
3. **LLaMA 2 and LLaMA 3** Ollama supported the local deployment of both LLaMA 2 and LLaMA 3, which played a critical role in our experimentation process. These models were fine-tuned on our curated mental health datasets to enhance their ability

to engage empathetically with users. A more detailed analysis of their performances is covered in a later section of the paper, but their seamless execution via Ollama was key in testing and deployment.

4. **Alternatives Considered** While Ollama was ultimately chosen for local model execution, alternative platforms were considered, including cloud-based services and custom Dockerized solutions. However, Ollama’s simplicity, resource efficiency, and privacy benefits made it the preferred option.
5. **Challenges and Future Potential** Despite the advantages, one limitation we faced was the need for higher computational power for more complex tasks. In future iterations, scaling up the system for larger deployments or running more resource-intensive versions of LLaMA may require a hybrid approach, integrating both local execution via Ollama and cloud-based services.

Overall, Ollama provided an accessible, flexible, and efficient framework for executing LLaMA models locally, and its role in the successful fine-tuning and deployment of LLaMA 2 and 3 for MindMate’s mental health-focused interactions was invaluable.

5 Core Component: Langchain

LangchainTeam (2023b) is a versatile and powerful framework specifically designed for building applications that leverage large language models (LLMs). It provides a structured, modular approach that simplifies working with LLMs, making it easier to integrate them into various applications. Langchain allows for the creation of complex conversational agents, dynamic tools, and knowledge-driven workflows by combining different components such as memory, chains, and tools.

For MindMate, Langchain is a natural fit because it facilitates the integration of multiple LLMs, each serving different purposes. For example, it supports combining general-purpose LLMs, like GPT, with models specifically trained on mental health data. This flexibility makes it easy to manage both the chatbot’s core conversations and its more specialized interactions, ensuring the system operates smoothly and efficiently.

Additionally, Langchain’s modularity helps streamline backend processes, such as handling conversation context, managing external data retrieval, and integrating other services like the Jitsi call feature, all within a single, cohesive framework.

Langchain is also useful for handling multi-step workflows, such as transitioning from chatbot interactions to live calls, which is a key component of MindMate. Its support for prompt templates and conversation memory helps maintain coherence and continuity in user interactions, creating a more natural and empathetic user experience—critical for mental health applications.

5.1 Langchain Functions Utilized

To build a highly functional mental health chatbot, we utilized several key functions from Langchain’s toolkit. These functions allowed us to load documents, create indexes for efficient retrieval, and embed textual data for interaction with the chatbot.

1. **TextLoader** The `TextLoader` function from `langchain.document_loaders` was employed to load text-based datasets into the system. This function allows the chatbot to read large amounts of text data efficiently, which is vital for its ability to provide relevant responses based on previous training data.
 - **Purpose:** The `TextLoader` serves as the initial point of data ingestion, enabling us to feed various text documents into the chatbot's memory for use in both training and real-time conversation generation.
 - **Application:** We used this function to load mental health-related textual datasets, such as therapy transcripts and self-help articles. This helped structure the chatbot's understanding of therapeutic conversations.
2. **VectorstoreIndexCreator** The `VectorstoreIndexCreator` from `langchain.indexes` was used to create an index of vector embeddings, which is a key step in optimizing the retrieval process. By converting text into vectors, the system can efficiently search for the most contextually relevant parts of the dataset during a conversation.
 - **Purpose:** This function aids in transforming raw textual data into a searchable format. It ensures that the chatbot can retrieve and present the most relevant information based on the user's input.
 - **Application:** `VectorstoreIndexCreator` was used to index our datasets, improving the chatbot's ability to provide coherent, context-aware responses in real time.
3. **OpenAIEmbeddings** `OpenAIEmbeddings` from `langchain_openai` was crucial for creating high-quality vector embeddings using OpenAI's language models. These embeddings serve as the foundation for the chatbot's ability to understand and process text at a deeper, semantic level.
 - **Purpose:** The `OpenAIEmbeddings` function allows for the transformation of input text into vectorized representations. This was crucial for understanding the contextual relationships between words, sentences, and larger text bodies.
 - **Application:** This function was used to create embeddings for the chatbot's mental health conversations, ensuring that user input could be processed effectively and generate accurate, empathetic responses.
4. **ChatOpenAI** The `ChatOpenAI` function from `langchain_openai` provides a simple interface to interact with OpenAI's chat models. It was implemented to facilitate seamless, back-and-forth interactions between users and the chatbot.
 - **Purpose:** The `ChatOpenAI` function integrates the chatbot with OpenAI's conversational models. This enabled real-time, high-quality dialogue generation in response to user queries.
 - **Application:** We employed `ChatOpenAI` to power the chatbot's conversational capabilities, allowing it to manage and respond to user queries in an empathetic, mental health-appropriate manner.

5.2 Conclusion

By combining these functions from Langchain, we were able to create a highly functional, context-aware chatbot. The efficient loading of datasets, creation of vectorized indexes, and integration with OpenAI’s language models provided the foundation for MindMate’s ability to offer real-time, empathetic mental health support.

6 Core Component: LLMs Used

Large Language Models (LLMs) have revolutionized the field of natural language processing (NLP) by enabling machines to understand and generate human-like text. These models, through advanced neural network architectures and vast datasets, have demonstrated remarkable capabilities in tasks ranging from language translation to conversational AI. For MindMate, the choice of LLMs was pivotal in ensuring that the chatbot could respond empathetically and accurately to user input, particularly in a mental health context where sensitivity and understanding are paramount.

In our project, we experimented with several LLMs to determine which would best align with our objectives for MindMate. Our selection included both general-purpose models and those fine-tuned for specific tasks. We focused on balancing conversational fluency with empathetic engagement, ultimately settling on a combination of models that could handle both general queries and specialized mental health-related discussions.

The models we utilized were chosen based on their performance in various domains. This included the open-source LLaMA series by Meta, specifically LLaMA 2 and 3, which offer a flexible and adaptable framework for building conversational agents. We also integrated GPT-3.5 and GPT-4o, part of OpenAI’s renowned GPT series, known for their extensive training on diverse datasets, making them particularly effective for more general conversational scenarios. Additionally, Gemini, developed as an advanced transformer-based model, was considered for its lightweight and resource-efficient properties, making it ideal for applications that required quick response times with limited computational resources.

Each model brought unique strengths to MindMate, and their integration into the system allowed for dynamic handling of user interactions. Depending on the type of input, the system could route conversations to the most appropriate LLM, ensuring that users received responses tailored to their needs—whether they sought general advice, emotional support, or simply a friendly conversation.

In the following subsections, we will delve into the individual LLMs we utilized in greater detail, outlining their architectures, key features, and how they contributed to the overall functionality of MindMate.

6.1 LLaMA 2

LLaMA 2 (Large Language Model Meta AI) is a state-of-the-art family of language models developed by Meta AI. Released in July 2023, LLaMA 2 builds on its predecessor, LLaMA, with improvements in architecture, training techniques, and dataset quality. These models are designed to generate human-like text based on prompts, making them highly versatile for

various natural language processing (NLP) tasks, including text generation, summarization, translation, and more.

6.1.1 KEY FEATURES OF LLaMA 2:

- **Diverse Model Sizes:** LLaMA 2 is available in multiple sizes (7B, 13B, and 70B parameters), allowing developers to choose a model that balances performance and resource requirements based on their specific needs.
- **Improved Fine-Tuning:** LLaMA 2 has been optimized for fine-tuning on specific tasks, enabling it to adapt effectively to niche applications like chatbots.
- **Robust Performance:** With enhanced training on diverse datasets, LLaMA 2 exhibits better comprehension and generation abilities, making it more contextually aware and responsive.
- **Open Access:** Meta AI has released LLaMA 2 under a more permissive license, facilitating research and commercial applications.

6.1.2 SUITABILITY FOR MENTAL HEALTH CHATBOT

When considering LLaMA 2 for building a chatbot focused on mental health, several factors contribute to its fit for this sensitive and impactful domain:

- **Natural Language Understanding:** LLaMA 2's advanced language understanding capabilities allow it to comprehend nuanced language and emotional tone, which is crucial when dealing with sensitive topics such as mental health.
- **Empathy and Responsiveness:** The model's ability to generate empathetic and supportive responses is vital for a mental health chatbot. By fine-tuning LLaMA 2 on mental health-related datasets, developers can enhance its capability to provide appropriate responses to users seeking help or guidance.
- **Customization:** LLaMA 2's fine-tuning capabilities enable developers to tailor the chatbot to specific mental health themes or issues, such as anxiety, depression, or stress management. This customization ensures that the chatbot can provide relevant resources and support.
- **Ethical Considerations:** LLaMA 2 allows developers to implement ethical guidelines and frameworks to ensure that the chatbot prioritizes user safety, provides accurate information, and encourages users to seek professional help when necessary.
- **Continuous Learning:** The model can be regularly updated with new information and findings in mental health research, ensuring that the chatbot remains current and informed.

6.2 LLaMA 3

LLaMA 3 (Large Language Model Meta AI) is the latest iteration of the LLaMA family of language models developed by Meta AI. Building upon the advancements of its predecessors, LLaMA 2 and the original LLaMA, this model incorporates enhanced architecture, improved training methodologies, and more extensive datasets. LLaMA 3 is designed to understand and generate human-like text, making it a versatile tool for a variety of natural language processing (NLP) tasks, including conversation, summarization, and context-aware dialogue.

6.2.1 KEY FEATURES OF LLAMA 3:

- **Enhanced Model Architecture:** LLaMA 3 features an optimized architecture that improves its ability to process complex language structures and maintain context over longer interactions.
- **Scalability:** Available in various sizes, LLaMA 3 supports different parameter counts, allowing developers to select a model that meets their performance and computational resource requirements.
- **Advanced Fine-Tuning:** The model is designed to facilitate fine-tuning, enabling developers to tailor it for specific applications, such as chatbots, by leveraging domain-specific datasets.
- **Robust Generalization:** With training on a broader and more diverse dataset, LLaMA 3 demonstrates superior generalization capabilities, making it effective at understanding various topics and contexts.
- **Open Source:** Meta AI continues its commitment to open research by making LLaMA 3 accessible under a permissive license, encouraging both academic and commercial use.

6.2.2 SUITABILITY FOR MENTAL HEALTH CHATBOT

LLaMA 3 is particularly well-suited for developing a chatbot focused on mental health for several reasons:

- **Natural Language Understanding:** LLaMA 3's advanced understanding of nuanced language and emotional expressions is crucial in the mental health domain, allowing the model to recognize and respond appropriately to users' feelings and concerns.
- **Empathy and Support:** The model can generate empathetic and compassionate responses, essential for a mental health chatbot. By fine-tuning LLaMA 3 on mental health-related conversations, developers can enhance its ability to provide comfort and support to users seeking help.
- **Customization and Flexibility:** The fine-tuning capabilities of LLaMA 3 allow developers to adapt the chatbot to address specific mental health issues, such as anxiety, depression, or stress management, ensuring relevant and personalized interactions.

- **Ethical Frameworks:** LLaMA 3 supports the implementation of ethical guidelines to ensure that the chatbot prioritizes user safety, provides accurate information, and encourages users to seek professional help when necessary.
- **Continuous Learning and Improvement:** The model can be updated with the latest research and findings in mental health, ensuring that the chatbot remains knowledgeable and can provide up-to-date resources and support.

6.2.3 CODE IMPLEMENTATION SUMMARY

The code begins by loading necessary environment variables from a `.env` file, including the API key for LangChain. This setup ensures secure access to the model's capabilities while enabling tracing for better debugging and performance monitoring.

6.2.4 PROMPT TEMPLATE

A prompt template is defined using the `ChatPromptTemplate` class from the LangChain library. This template specifies how the model should respond to user queries, ensuring that responses are concise and relevant. The template includes a system message to guide the model and a user message formatted to include the user's question.

6.2.5 LLM INITIALIZATION

The Ollama LLM is initialized with the `llama2` model, preparing it for subsequent queries. An output parser, `StrOutputParser`, is also instantiated to handle the responses generated by the model.

6.2.6 DATA LOADING

The application includes a function to load conversation data from a specified text file. This function reads and strips lines from the file, returning a list of valid conversation entries, which will be used for question matching.

6.2.7 DATASET CHECK FUNCTIONALITY

A dedicated function checks if a user's question exists in the loaded dataset. If a match is found, the application returns a predefined response indicating that the answer is available in the dataset. This feature enhances the chatbot's ability to provide quick responses for frequently asked questions.

6.2.8 WEB ROUTES

Flask routes are established to serve the main page and the chat interface. The `/query` endpoint processes user input through a POST request. It handles the query by checking the dataset and, if the question is not found, invoking the Ollama LLM to generate a response based on the prompt template.

6.2.9 ERROR HANDLING

The query function includes error handling to capture and report any exceptions that occur during the processing of user input. This ensures a smooth user experience by providing feedback in case of issues.

6.2.10 RUNNING THE APPLICATION

The application is configured to run in debug mode on port 5001, making it accessible for users to interact with the chatbot through a web interface.

6.3 GPT 4o

GPT-4 (Generative Pre-trained Transformer 4) is an advanced language model developed by OpenAI. Released in March 2023, GPT-4 builds on the capabilities of its predecessors (GPT-3 and GPT-3.5) with significant improvements in understanding context, generating coherent text, and performing complex tasks. As a transformer-based model, GPT-4 utilizes deep learning techniques to understand and produce human-like text based on the input it receives. It has been trained on a diverse range of data sources, which contributes to its versatility across various applications.

6.3.1 KEY FEATURES OF GPT-4:

- **Enhanced Language Understanding:** GPT-4 exhibits a deeper understanding of context and subtleties in language, allowing it to generate more accurate and relevant responses.
- **Multimodal Capabilities:** Unlike its predecessors, GPT-4 can process both text and images, enabling it to understand and respond to a wider variety of inputs.
- **Improved Fine-Tuning:** GPT-4 supports more effective fine-tuning, allowing developers to tailor the model for specific applications and domains, including mental health.
- **Robust Performance:** With increased model size and improved algorithms, GPT-4 demonstrates higher accuracy, coherence, and context retention, making it suitable for complex conversational scenarios.
- **Ethical Considerations and Safety Features:** OpenAI has implemented various safety measures and ethical guidelines to minimize harmful outputs and ensure responsible use of the model.

6.3.2 SUITABILITY FOR MENTAL HEALTH CHATBOT

GPT-4 is particularly well-suited for developing a chatbot focused on mental health for several reasons:

- **Natural Language Processing:** The advanced language understanding capabilities of GPT-4 enable it to comprehend nuanced emotional language, making it effective in

recognizing and addressing the feelings and concerns of users seeking mental health support.

- **Empathy and Support:** GPT-4 is capable of generating empathetic and supportive responses, which are essential for a mental health chatbot. By fine-tuning the model with mental health-specific data, developers can enhance its ability to offer comfort and guidance.
- **Customization and Personalization:** The model can be fine-tuned to focus on specific mental health topics (such as anxiety, depression, or stress management), allowing the chatbot to provide tailored and relevant responses based on user interactions.
- **Engagement in Meaningful Conversations:** GPT-4's ability to maintain context over longer conversations enables it to engage users in meaningful discussions, helping them explore their thoughts and feelings more deeply.
- **Ethical Frameworks:** OpenAI has prioritized safety and ethical considerations, allowing developers to implement guidelines to ensure the chatbot operates responsibly. This includes encouraging users to seek professional help when necessary and providing accurate information.
- **Continuous Learning and Adaptation:** Although GPT-4 itself does not learn from individual interactions, developers can update the model with new mental health research and findings to ensure that the chatbot remains informed and provides up-to-date support.

6.3.3 CODE IMPLEMENTATION SUMMARY

The implemented code leverages Flask to create a web application that acts as a mental health chatbot using the GPT-4 model.

1. **Environment Setup** The code begins by loading necessary environment variables, including the OpenAI API key. This key is essential for making API calls to the GPT-4 model and ensures secure access to the model's capabilities.
2. **Data Loading** It utilizes the `TextLoader` from the LangChain library to load conversation datasets and prompts. This functionality allows the chatbot to reference past interactions and use contextually relevant prompts, enhancing the quality and relevance of its responses.
3. **Embedding Creation** OpenAI's embeddings are created to facilitate efficient information retrieval, which is crucial for providing accurate and contextually appropriate responses based on user queries. These embeddings help in understanding user intents and matching them with the relevant dataset.
4. **Query Function** The query function is responsible for processing user input. It receives queries through the web interface and sends them to the GPT-4 model using the LangChain framework. The function formats the query, invokes the model, and

retrieves the response. This response is then returned to the user in a structured JSON format, allowing for a seamless conversational experience.

6.4 Gemini

Gemini, developed by Google DeepMind, is a state-of-the-art language model that excels in various natural language processing (NLP) tasks. Its robust capabilities make it particularly well-suited for applications in mental health support, where sensitivity and responsiveness are paramount.

6.5 Key Capabilities

The model's suitability for a mental health chatbot can be attributed to several key capabilities:

- **Contextual Understanding:** Gemini can understand and maintain context across conversations, allowing for coherent and relevant interactions.
- **Emotional Sensitivity:** The model generates responses that are emotionally resonant, which is critical in addressing the nuanced emotional states of users.
- **Adaptability:** Gemini can process both structured and unstructured data, enabling it to handle diverse user expressions and queries effectively.

6.6 Importance in Mental Health Settings

The nature of mental health conversations demands specific qualities from a language model:

- **Empathy and Validation:** Users often seek not just information but emotional support and validation. Gemini's training enables it to respond in ways that acknowledge and validate user feelings.
- **Therapeutic Alignment:** The model is designed to provide responses aligned with therapeutic goals, offering coping strategies and insights when necessary.

6.7 Flexibility in Conversations

Gemini's ability to adapt its responses is particularly beneficial for mental health applications:

- **Personalized Interactions:** The model tailors its dialogue to the unique needs of individual users, enhancing the overall experience.
- **Open-Ended Conversations:** It effectively manages open-ended questions, ensuring that discussions remain engaging and meaningful.

6.8 Code Implementation

The integration of the Gemini model within the MindMate chatbot is achieved through a well-structured codebase that facilitates efficient interaction with a custom dataset. The following outlines the key components of the implementation:

1. **Environment Configuration:** The code begins by importing essential libraries, including `langchain` for document management and embedding storage, as well as the `dotenv` module for secure handling of environment variables. This includes the Google API key necessary for accessing the Gemini model.
2. **Dataset Handling:** A custom function, `load_custom_dataset`, is implemented to read text files containing conversation data. This function employs error handling to manage potential encoding issues, ensuring that the data is correctly formatted as `Document` objects for subsequent processing.
3. **Embedding Generation:** The `create_embeddings` function utilizes a Hugging Face model, specifically `sentence-transformers/all-MiniLM-L6-v2`, to convert text data into vector embeddings. These embeddings are stored in a FAISS vector store, which enables efficient similarity searches for retrieving relevant responses.
4. **Setting Up the QA Chain:** A retrieval-based QA chain is established through the `setup_qa_chain` function. This function connects the vector store to the Gemini model, allowing the chatbot to query the embedded dataset and generate contextually appropriate responses.
5. **Execution Workflow:** The main block of the script orchestrates the loading of the dataset, embedding creation, and setup of the QA chain. It prompts the user for input and retrieves answers using the trained model, with robust error handling to manage any potential issues during interaction.

This code structure not only optimizes the performance of the Gemini model but also ensures a responsive and engaging user experience, vital for mental health applications.

7 Core Component: Call Feature

7.1 Description

The Call Feature in MindMate is a critical component, designed to seamlessly bridge the gap between the chatbot's automated responses and human interaction. While MindMate can handle many mental health-related conversations with empathy and understanding, there are situations where a user may feel the need to speak to a real person. The call feature allows users to initiate a direct conversation with a licensed therapist or a highly qualified mental health professional.

Users can access this feature via a simple "Call" button embedded in the chatbot's interface. Once the user clicks the button, a real-time video or voice call is initiated through the integration of the Jitsi API, a secure, open-source video conferencing solution. Users are assigned to virtual meeting rooms in a round-robin manner, ensuring that they are connected to an available therapist or mental health professional based on their position in the queue.

7.2 How the Call Feature Works

The Call Feature in MindMate is designed to enhance the user experience by allowing users to transition from text-based interactions to a real-time conversation with a mental health

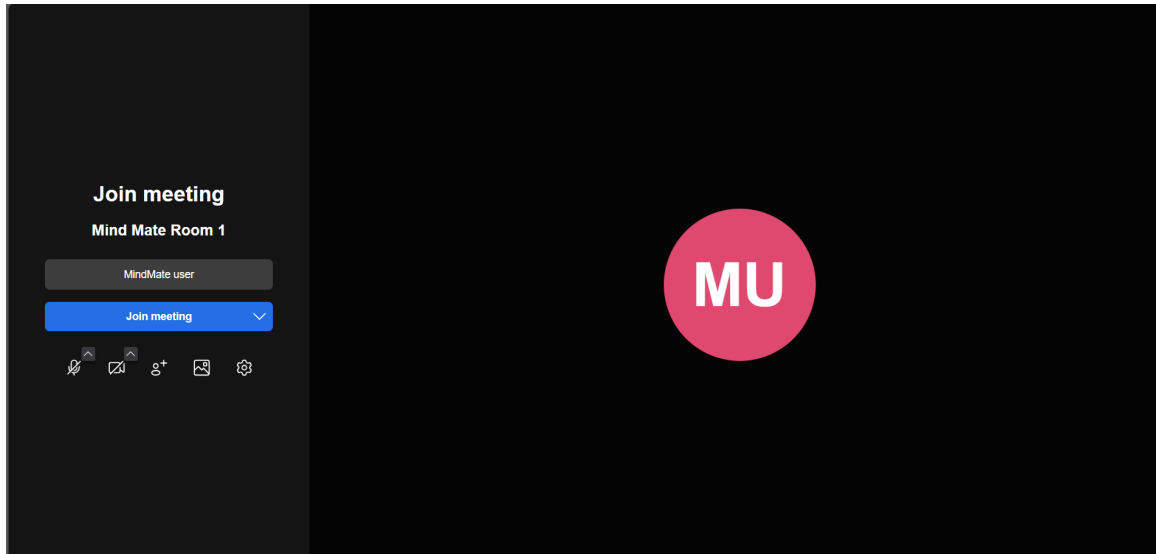


Figure 1: Jitsi call joining screen

professional. However, this feature is not available from the start of the conversation; it becomes active after a certain amount of time has elapsed or if the conversation has been ongoing for a while. This design choice ensures that users have adequate time to express their thoughts and feelings to the chatbot before seeking direct assistance.

Once the call feature is activated, users can see the “Call” button prominently displayed in the interface. When they click this button, the system initiates a series of actions:

1. **Notification to the Moderator:** Upon activation of the call feature, an email is sent to the assigned moderator by one of our extra email IDs, notifying them to join one of the virtual meeting rooms. This email includes the meeting link and any relevant meeting room number.
2. **Room Assignment:** MindMate features multiple meeting rooms (Room 1 through Room 5) for connecting users to therapists. The system employs a round-robin assignment logic, ensuring that users are assigned to different rooms based on their order of button clicks. The first time a user clicks the call button, they are connected to Room 1.

The second click connects them to Room 2, and so on. After the fifth click, the system cycles back, connecting the sixth click to Room 1 and the seventh click to Room 2, continuing this pattern. This setup allows for balanced utilization of available therapists across the rooms.

3. **Delay for Moderator Joining:** To create a seamless user experience, there is a brief delay after the user clicks the call button. This delay allows the moderator to join the meeting room first, ensuring that when the user enters, they are greeted by a professional and can begin their conversation without waiting unnecessarily.

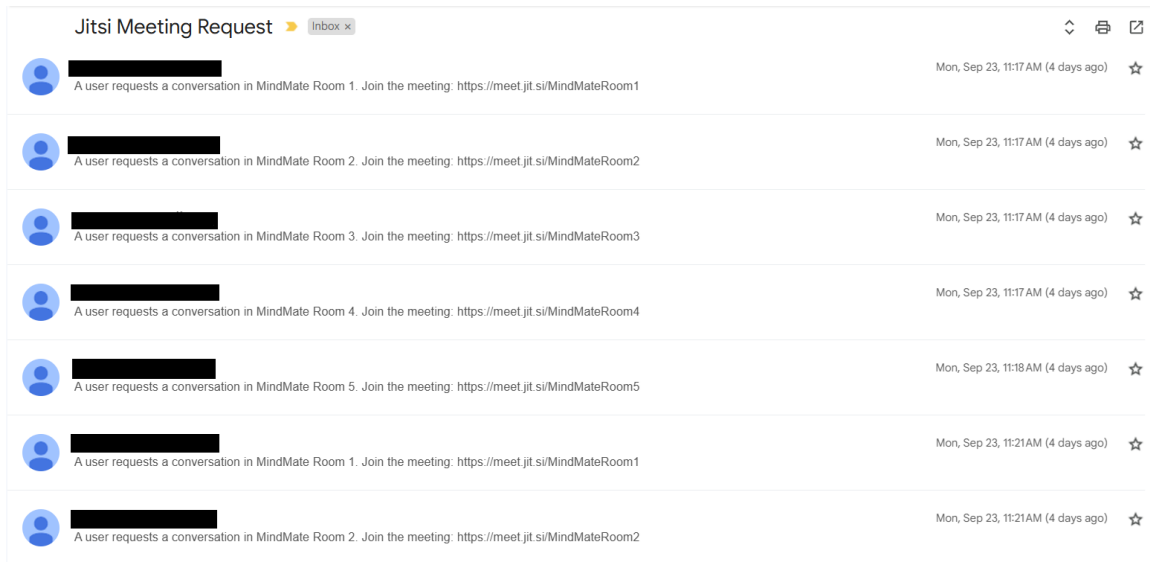


Figure 2: Mails received by moderator with mail Id censored for privacy

4. **User Feedback:** During this delay, users receive a message that says, "Wait a few seconds while the moderator joins the call." This feedback helps manage user expectations and reduces anxiety as they await their connection to a therapist.

By structuring the call feature in this manner, MindMate ensures that users can transition to a live conversation effectively and confidently, providing a crucial element of support when needed most.

7.3 Evaluation

The call feature is both a functional and user-facing component, requiring extensive testing to ensure reliability, ease of use, and effectiveness in providing real-time mental health support.

1. Functionality Testing

The call feature's functionality was tested by integrating the Jitsi API to initiate video or voice calls and ensuring smooth transitions from chatbot interactions to live conversations. Testing involved validating that users were consistently assigned to the correct virtual rooms and that therapists received notifications via email, containing the relevant meeting link.

Multiple test scenarios were run to ensure that the correct therapist was notified based on the room assignment logic (round-robin allocation). The functionality was further validated by testing under various conditions, such as network fluctuations and multiple simultaneous users, to ensure the reliability and stability of the call feature.

2. User Testing

While broad user testing was not conducted, we engaged quite a lot of colleagues to

evaluate the user experience of the call feature. Feedback focused on ease of access and the intuitiveness of the interface.

Early tests demonstrated that users found it straightforward to initiate calls and that therapists could quickly join the virtual room with minimal delay. Additionally, the 15-second cooldown timer following the call button press ensured that accidental double clicks were prevented, reducing confusion for both users and therapists.

Every tester provided the importance of clear feedback to users while waiting for the therapist to join. As a result, a message was incorporated to inform users to "Wait a few seconds while the moderator joins the call," helping to reduce anxiety during the brief waiting period.

3. Future Testing and Scalability

To ensure the scalability of the call feature, future testing will focus on how the system performs with a larger number of users and therapists. Load testing will be conducted to measure the system's ability to handle peak traffic and ensure that no user experiences a long delay in connecting to a professional.

Furthermore, incorporating user feedback from broader testing will be key in refining the user interface and improving the overall functionality of this feature.

7.4 Alternative Services Considered

While implementing the call feature, several other video conferencing services were evaluated, including Telegramtel (2024), MatrixMatrix.org (2023), and Jitsi. Each service offered unique advantages and considerations:

1. **Telegram:** Although Telegram provides voice and video call functionalities, its primary use case centers around messaging and group chats. Integrating Telegram would have required additional steps for users, such as downloading the app or creating an account, which could potentially deter users from utilizing the call feature.
2. **Matrix:** Matrix is a decentralized communication protocol that offers rich messaging capabilities and support for voice and video calls. However, its implementation can be complex and may not be as user-friendly for those unfamiliar with its setup. Given that MindMate aims for accessibility, we sought a solution that would minimize barriers for users.
3. **Jitsi:** Ultimately, we chose Jitsi for its open-source nature, ease of integration, and user-friendly interface. Jitsi allows for seamless video conferencing without the need for user accounts or downloads, ensuring a frictionless experience for users seeking mental health support.

Additionally, its emphasis on privacy and security aligns with our commitment to protecting users' sensitive information. Jitsi's features, such as screen sharing and recording options, further enhance the potential for interactive sessions between users and therapists.

By selecting Jitsi as our video conferencing solution, we aim to provide a reliable and user-friendly call feature that meets the needs of MindMate users while maintaining a focus on security and accessibility.

8 Self-Assessment

8.1 How would you judge the final product?

We judge the final product to be quite successful, as it aligns closely with our initial expectations while also surpassing some of them. Throughout the development process, we made several adjustments based on user feedback, which played a crucial role in refining the product. The iterative approach we took allowed us to adapt to real-world usage and implement improvements that enhanced overall functionality. The end result is a product that not only meets our objectives but also resonates well with our target audience.

One of the notable strengths of the product is its flexibility. Rather than relying solely on a single, well-known large language model (LLM), we undertook a comprehensive evaluation of three different models. This comparative analysis provided us with valuable insights into each model's strengths and weaknesses, enabling us to select the one that performed best under various conditions. The resulting blend of capabilities from the different models contributed to a more robust and effective solution, ensuring users receive high-quality interactions.

Additionally, we implemented a significant feature that facilitates call interactions after extended conversations with the chatbot. This enhancement allows for a more personal touch and serves as a vital bridge between automated support and human intervention. We also prioritized transparency in our design by ensuring that the chatbot clearly communicates its non-human nature, requiring users to agree to terms and conditions. This commitment to clarity is crucial for establishing trust and preventing misunderstandings, which is especially important in a mental health context.

Our rigorous testing protocols further confirmed that the bot consistently avoided producing rude or misleading messages, which was a vital aspect of the project. Instead of offering concrete advice, the chatbot provides gentle, supportive guidance designed to keep conversations flowing and subtly distract users from their challenges. This design emphasizes a conversational AI experience that is both engaging and professional, ensuring users feel supported throughout their interactions. Overall, the product's design reflects a thoughtful balance of functionality, user engagement, and ethical considerations.

8.2 Is it useful

The product offers substantial utility by providing individuals in mental distress with a secure and supportive space to converse. Unlike conventional chatbots, our system prioritizes creating an environment where users can express themselves freely, without the fear of inappropriate responses or breaches of trust. This sense of security is particularly crucial for individuals who may be hesitant to seek professional help immediately, making the chatbot a valuable initial step toward emotional support. By fostering a safe and non-judgmental atmosphere, we empower users to engage openly with their feelings, which is essential for

mental well-being.

One of the standout features of our product is its seamless transition to professional help after prolonged interaction with the chatbot. This functionality ensures that users can receive timely assistance when they need it most, circumventing the long wait times often associated with traditional therapy appointments. By facilitating earlier intervention, we not only alleviate user anxiety but also help reduce the burden on mental health services, allowing professionals to focus on those in critical need. This direct line to professional care enhances the overall user experience and ensures that individuals are not left feeling isolated during their time of distress.

User testing revealed that conversations with the chatbot flowed smoothly and encouragingly, significantly enhancing user engagement and effectiveness. Feedback from participants indicated that the system maintained a supportive tone throughout interactions, which is essential for building trust and ensuring a constructive experience. Users expressed appreciation for the conversational style, which helped them feel more comfortable discussing sensitive topics. This supportive approach is crucial for individuals navigating their mental health challenges and reinforces the product’s utility as a primary support tool.

Overall, our product effectively addresses a significant gap in mental health care by providing immediate and reliable support, while also offering the option to connect with professionals when necessary. Its design not only prioritizes user safety and comfort but also seeks to streamline access to mental health resources. By bridging the gap between automated support and human intervention, we believe this product has the potential to make a meaningful impact on mental health care accessibility, ultimately fostering a healthier, more supportive environment for those in need.

8.3 Is it efficient?

The product is engineered for efficiency, both computationally and in terms of user workflow. From a computational perspective, we optimized the underlying algorithms to ensure the system operates smoothly and swiftly, even when managing multiple conversations concurrently. This optimization minimizes response times, allowing users to engage with the chatbot without noticeable delays. Additionally, the resource usage has been kept manageable, ensuring scalability for broader usage without excessive computational demands.

In terms of workflow efficiency, the chatbot dramatically reduces the time users spend seeking help. Instead of enduring long waits for therapy appointments, users can immediately engage in meaningful conversations with the chatbot. This instant interaction provides relief much quicker than conventional mental health support methods. The streamlined process—from engaging with the chatbot to potentially connecting with a professional—saves users valuable time and promotes earlier interventions.

Furthermore, the chatbot’s ability to comprehend and respond to user queries effectively enhances the overall user experience. Users do not need to repeatedly clarify their concerns, as the system is designed to maintain smooth conversational flow. This design reduces frustration, ensuring users feel supported without unnecessary delays. In summary, the

product achieves high efficiency in both system performance and user experience, making it a practical tool for individuals seeking immediate assistance.

8.4 What is working well?

One of the standout features functioning exceptionally well is the call function, which allows users to seamlessly connect with specialized professionals after engaging with the chatbot. This feature is particularly valuable because it ensures that users can transition effortlessly from automated interactions to live conversations with experts when they require additional support. The reliability of this function significantly enhances the user experience, providing timely access to professional help that is crucial for individuals facing immediate challenges. By offering a direct line to professional care, we can effectively address users' needs while ensuring they do not feel abandoned after their initial interaction with the chatbot.

Moreover, the chatbot excels in maintaining a natural and supportive conversation flow throughout user interactions. During testing, we observed that the system effectively engaged users by offering helpful, non-intrusive suggestions that advanced the conversation without feeling forced or repetitive. This ability to provide contextually relevant responses not only keeps the dialogue engaging but also fosters a sense of companionship for users. Participants reported that interacting with the chatbot felt easy and intuitive, which is essential for building a supportive environment where users feel comfortable sharing their thoughts and feelings.

Additionally, the design of the chatbot promotes trust and engagement, ensuring that users remain involved throughout their conversations. We also implemented a calming music feature that plays soothing melodies during interactions, helping users relax and fostering a positive, tranquil environment. This auditory element not only enhances the overall user experience but also creates a more inviting atmosphere for sharing personal thoughts and feelings. By combining supportive dialogue with calming music, we cultivate a sense of safety and comfort, which is vital for individuals seeking mental health support. This holistic approach encourages users to open up more freely, reinforcing their engagement and trust in the system.

8.5 What is missing?

Despite the product's strengths, there are several features we could not fully implement due to time constraints and technical challenges. One valuable addition would be training the model with a more positive dataset to enhance its ability to provide uplifting responses. Moreover, a mechanism to detect triggering words or phrases within conversations could significantly improve user safety. Implementing such a feature would allow the chatbot to recognize distress signals, escalating conversations to professionals if certain keywords are repeated, ensuring immediate support when it is most needed.

Further refinement of the conversation flow is also necessary to enhance user engagement. Although the current system functions well, adopting a more casual and friendly conversational style could create a more inviting atmosphere, encouraging users to share more

openly. This adjustment, in conjunction with the proposed safety features, would significantly enhance the overall effectiveness and user experience of the chatbot.

8.6 How could it be turned into a real product?

To transform this project into a viable product, the initial focus should be on refining the underlying model to ensure its performance is exceptional. This involves rigorous testing and optimization of the chatbot’s responses, ensuring it can effectively handle a wide range of user inquiries. Expanding the training dataset with diverse and positive conversational examples will further enhance the bot’s ability to deliver supportive and contextually relevant responses. By prioritizing these improvements, we can create a system that not only meets but exceeds user expectations, positioning it as a leader in the mental health chatbot space.

In addition to refining the model, navigating the legal and regulatory landscape is crucial for establishing a legitimate product. This includes completing the business registration process (Gewerbeanmeldung) and ensuring compliance with data protection laws, such as the General Data Protection Regulation (GDPR). Implementing robust data security measures to protect user information is essential for building trust with our audience and operating within the healthcare domain. Engaging with legal experts to guide us through these processes will ensure we meet all regulatory requirements. Consulting with professionals experienced in the German healthcare system can further streamline our efforts. Their expertise can help us navigate the complexities of healthcare regulations related to telehealth services. Additionally, forming partnerships with existing mental health organizations and practitioners can enhance our credibility and resource availability, supporting our outreach initiatives.

To maximize our product’s impact, we also aim to develop a comprehensive database for securely storing user information. This database will not only support the chatbot’s functionality but also enable us to analyze user interactions, informing future feature development. By leveraging this data and incorporating user feedback mechanisms, we can continuously improve the user experience and introduce new functionalities that address our audience’s evolving needs.

A critical aspect of our vision is to engage with government bodies for potential funding to provide this app free of charge to users. By collaborating with therapists who can offer video consultations, we aim to eliminate common barriers to accessing mental health support, such as long wait times and the difficulty of finding qualified professionals. This initiative not only facilitates faster access to care but also promotes mental well-being across the population. With initial success in Germany, we aspire to replicate this model in other countries, ultimately striving for a global reach. Our overarching goal is to ensure that mental health support is accessible to everyone, fostering a healthier, more peaceful world.

8.7 Should it?

The question of whether this product should be brought to market is both timely and highly relevant, especially in light of the current mental health crisis affecting Germany and much

of Europe. During a recent business workshop, we engaged with various stakeholders who emphasized the critical challenges individuals face in accessing mental health care. Many participants shared their frustrations about the significant shortage of available therapy appointments, resulting in long wait times that leave individuals feeling unsupported and isolated. This scarcity of resources is particularly concerning in a landscape where mental health issues are on the rise, making it essential to find innovative solutions to bridge this gap.

Given this context, launching our product could provide a much-needed resource for individuals seeking immediate mental health support. By leveraging technology, we can offer an accessible platform that allows users to engage in conversations at their own pace and comfort level. Our chatbot is designed to provide immediate assistance, allowing individuals to voice their concerns without the intimidation that often accompanies traditional therapy settings. This approach not only empowers users to take the first step towards seeking help but also alleviates the pressure on mental health services by acting as a valuable intermediary.

Furthermore, the integration of features that facilitate a smooth transition to professional help ensures that users are not left to navigate their challenges alone. By providing timely access to experts when needed, our product can effectively complement existing mental health services rather than replace them. This dual approach allows us to address both immediate needs and long-term mental health goals, creating a comprehensive support system that can adapt to users' varying levels of distress.

Ultimately, introducing this product to the market aligns with a growing recognition of the importance of mental health awareness and support. In a time when individuals are increasingly seeking ways to manage their emotional well-being, our solution can serve as a vital tool in fostering mental resilience. By making mental health support more accessible and engaging, we can contribute to a larger societal shift towards prioritizing mental well-being and ultimately improve the quality of life for countless individuals across the region.

9 User Interface Implementation of MindMate

The MindMate application serves as a digital companion aimed at providing mental health support through an interactive chat interface. Given the pressing need for accessible mental health resources, especially during the ongoing crisis in Germany and Europe, the design of MindMate focuses on ease of use, user engagement, and emotional comfort. This section will explore the UI implemented in two primary HTML files: `opening.html` and `chat.html`. Each component is meticulously designed to enhance the user experience and facilitate a supportive environment.

9.1 Overview of the User Interface

9.1.1 DESIGN PHILOSOPHY

The design philosophy behind MindMate revolves around creating a calm and inviting atmosphere. The color palette, font choices, and overall layout work harmoniously to evoke

feelings of safety and openness. Recognizing that many users may approach the app during vulnerable moments, the UI aims to minimize anxiety and foster a sense of belonging.

9.1.2 TARGET AUDIENCE

MindMate is designed primarily for individuals seeking mental health support, particularly those facing challenges accessing traditional therapy methods. This includes young adults, students, and professionals who may benefit from an immediate, anonymous, and non-judgmental space to express their thoughts and feelings.

9.1.3 OPENING PAGE (OPENING.HTML)

The opening page acts as the first point of contact for users, setting the stage for their interaction with the MindMate application.

1. Layout and Structure

- **Straightforward Layout:** The layout of `opening.html` is intuitive, ensuring users can easily navigate the interface.
- **Flexbox Structure:** The use of CSS Flexbox properties allows the elements to be centred both horizontally and vertically, creating a visually appealing and accessible design. The main container holds both the image and content, making it easy for users to engage with the page.
- **Responsive Design:** The page is designed to be fully responsive, adapting to different screen sizes. This adaptability is crucial for accessibility, ensuring users on mobile devices have an equally engaging experience.

2. Visual Elements

- **Background Color:** The soft gray background (`#f0f0f0`) offers a calming effect, which is crucial for mental health applications. This neutrality allows the content to stand out without overwhelming the user.
- **Container Styling:** The white container with rounded corners and padding creates a sense of comfort and focus. It acts as a visual "bubble," shielding users from distractions outside the app.

3. Branding and Messaging

- **Logo and Imagery:** The logo displayed on the opening page (sourced from the static images) reinforces brand recognition. A well-designed logo can foster trust and familiarity among users.
- **Main Title and Subtitle:** The title "*MindMate*" is rendered in a bold, cursive font, imparting a sense of warmth and companionship. The subtitle "*Your Personal Wellness Companion*" succinctly communicates the app's purpose.

4. Call-to-Action Button

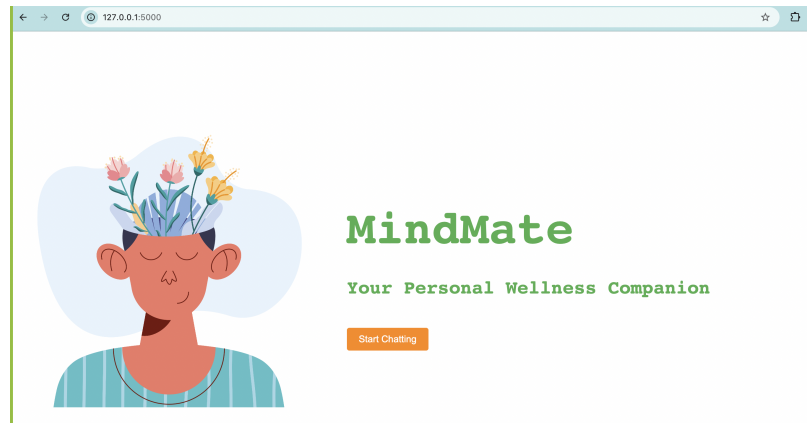


Figure 3: Opening.html

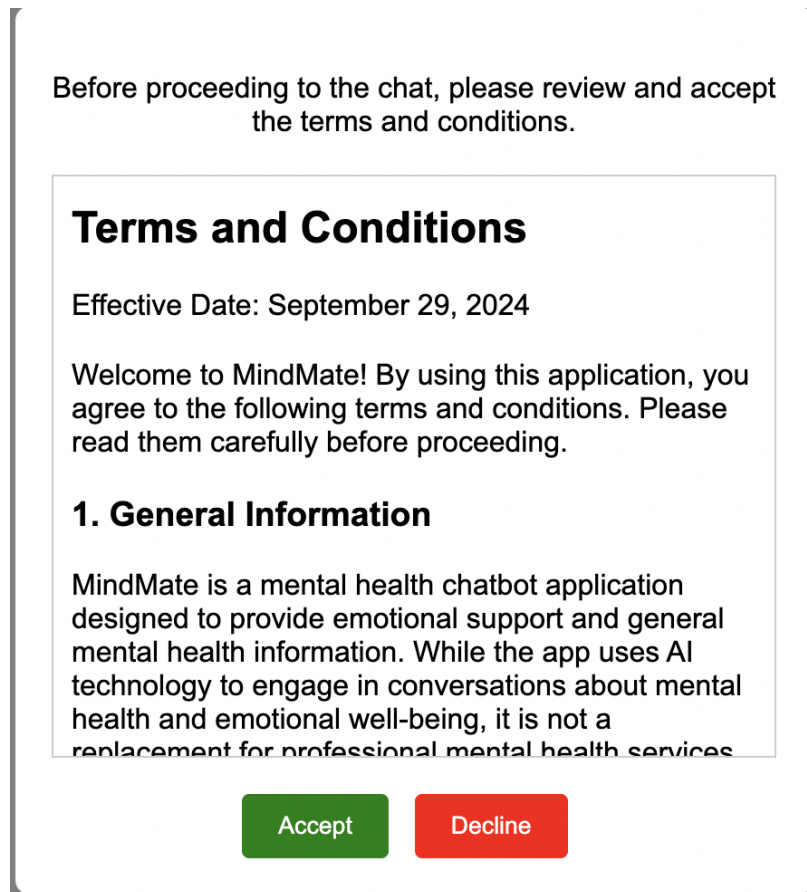


Figure 4: Terms and Conditions

- **Design and Placement:** The "Start Chatting" button is strategically placed in the center of the page, serving as the primary call to action. Its vibrant orange

hue (hsl(32, 100%, 50%)) contrasts sharply with the background, drawing the user's eye and encouraging interaction.

- **Functionality:** Upon clicking the button, a modal appears, prompting users to accept the terms and conditions. This feature not only facilitates legal compliance but also establishes transparency regarding user expectations and responsibilities.

5. Modal Window for Terms and Conditions

- **Accessibility and Visibility:** The modal window is designed to cover the entire viewport, dimming the background to ensure users focus on the terms and conditions. This approach minimizes distractions and emphasizes the importance of the information presented.
- **Content Structure:** The terms are presented in a scrollable format, allowing users to review the entire content without feeling overwhelmed. By providing this information upfront, users can make informed decisions about their participation in the chat.
- **Action Buttons:**
 - **Accept Button:** Allows users to confirm their understanding and agreement, facilitating a smooth transition to the chat interface.
 - **Decline Button:** Gives users the option to exit the modal without proceeding, respecting their autonomy and comfort level.

9.2 Chat Interface (chat.html)

The chat interface is where users engage with the chatbot, making it the most crucial component of the MindMate application.

9.2.1 STRUCTURE AND LAYOUT

- **Container Design:** The chat interface is organized within a well-defined container that houses all the interactive elements. This structure promotes a clean and organized look, helping users focus on the conversation.
- **Scroll Behavior:** The chat container is designed to be scrollable, allowing users to review past messages easily. This feature is particularly beneficial in long conversations, ensuring users can always refer back to previous exchanges without losing context.

9.2.2 HEADER SECTION

- **Title and Branding:** The header contains the app logo and the title "*Mind Mate*," reinforcing brand identity while providing a welcoming introduction to the chat.
- **Consistency:** The color scheme of the header matches the opening page, ensuring a cohesive experience as users navigate through the application.

9.2.3 CHAT FUNCTIONALITY

- **Message Display:** The chat messages are differentiated by color to signify the speaker. User messages appear in a muted gray, while bot messages are displayed in a soothing green. This colour-coding helps users quickly identify who is speaking and enhances the overall readability of the conversation.
- **Message Styling:** Each message is styled with rounded corners and padding, contributing to a friendly and inviting appearance. This design choice helps create a comfortable environment where users can freely express themselves.

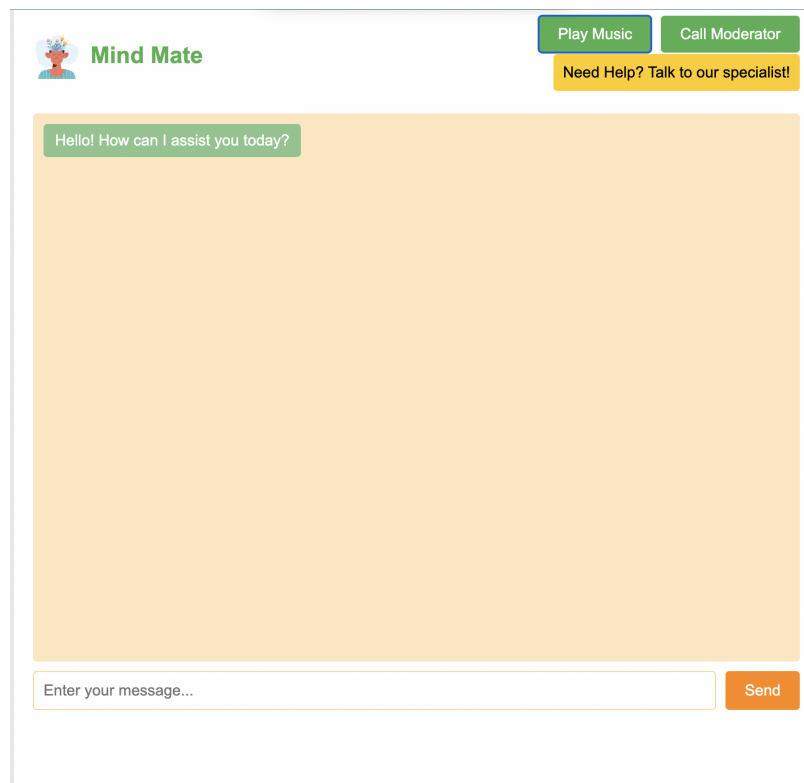


Figure 5: chat.html

9.2.4 MESSAGE INPUT FORM

- **User Interaction:** The message input form is straightforward and user-friendly. The input field encourages users to type their thoughts and feelings while providing a clear placeholder ("*Enter your message...*") that guides them.
- **Submission Mechanism:** The "Send" button allows users to submit their messages with ease. Upon clicking, the message is appended to the chat container, ensuring the conversation feels fluid and dynamic.

9.2.5 AUDIO FEATURES

- **Background Music:** An audio element is embedded in the chat interface, playing calming piano music upon entry. This feature is designed to create a serene environment that enhances user comfort during conversations.
- **Play/Pause Functionality:** The "Pause Music" button provides users with control over their auditory experience. This feature respects user preferences, allowing them to choose whether or not to have music playing during their chat session.

9.2.6 HELP AND SUPPORT FEATURES

- **Call Moderator Button:** This button is disabled upon entry, only becoming active after a certain time period, indicating to users that professional help is available if they need it. When clicked, it initiates a call to a moderator, ensuring users can reach out for immediate support in critical moments. This functionality is vital for creating a safety net for users who may feel overwhelmed during their interactions.
- **Help Message Popup:** A small help message appears in the top right corner, offering guidance on seeking additional support. This message serves as a gentle nudge, reminding users that help is available and that they are not alone in their journey.

9.2.7 ERROR HANDLING AND USER FEEDBACK

- **User Feedback Mechanism:** The application includes error handling features that display messages in the chat container if there are issues during message submission. This proactive approach reassures users that their concerns are acknowledged and mitigates frustration during interactions.
- **Automatic Scrolling:** The chat interface is designed to scroll to the latest message automatically, ensuring users are always aware of the most recent interactions. This feature enhances usability, especially during fast-paced conversations.

9.3 Overall User Experience

9.3.1 USER-CENTRIC DESIGN

- **Design Priorities:** The design of *MindMate* prioritizes user comfort, engagement, and ease of use. Every element, from the inviting color palette to the intuitive navigation, contributes to a positive experience for users seeking mental health support.
- **Emotional Considerations:** By employing soothing colors, gentle typography, and thoughtful interactions, *MindMate* creates an emotionally supportive environment. This approach is crucial, especially considering the sensitivity of the subject matter.

9.3.2 ACCESSIBILITY AND INCLUSIVITY

- **Responsive Design:** The application adapts seamlessly to various screen sizes, catering to users on smartphones, tablets, and desktops. This inclusivity ensures that anyone can access mental health support regardless of their device.

- **Font Size and Color Contrast:** The choice of font size and contrast ensures readability for users with visual impairments, further promoting inclusivity in accessing mental health resources.

10 User Survey

The MindMate user review survey using google forms LLC (2023) is designed to comprehensively assess the app's performance, usability, and overall impact on users' mental health experiences. The first few questions focus on the ease of interaction and the app's effectiveness in addressing users' concerns.

By asking users to rate how simple it was to navigate MindMate and how helpful they found its features, the survey aims to gauge whether the app delivers on its promise of providing accessible mental health support. These questions not only help in understanding the app's usability but also provide insights into how well the app addresses core mental health concerns, which is crucial for ensuring it serves its intended purpose.

Furthermore, the survey takes into account the emotional response of users by asking how supported they felt after engaging with MindMate. This aspect is essential, as the feeling of being emotionally supported can play a significant role in mental health outcomes. The survey also explores users' likelihood of returning to the app in the future, which is an important metric for assessing user satisfaction and long-term engagement.

Additionally, it dives into potential discomforts or difficulties users might have faced, encouraging honest feedback about any challenges they encountered. These questions help in identifying areas for improvement, ensuring that the app remains a safe and comfortable space for all users.

Lastly, the survey gathers feedback on various design and feature elements, such as the user interface (UI) and the in-app music designed to promote relaxation. The questions regarding the UI help gauge how intuitive and aesthetically pleasing the app is, while those about the music explore its effectiveness in creating a calming environment. There's also a focus on the call feature, asking how seamless the transition was when users opted to speak with a professional, ensuring that the app's integration of live support works smoothly.

The survey concludes with an open-ended question, allowing users to provide any additional feedback, ensuring that no aspect of their experience is left unaddressed. This feedback will be instrumental in helping the MindMate team make iterative improvements, ensuring the app continues to meet users' needs and deliver a high-quality mental health support experience.

User Review Survey

Please rate MindMate

swati.selvaraj11@gmail.com [Switch account](#)

Not shared

1. How easy was it to interact with MindMate?

1 2 3 4 5

Very easy ☐ ☐ ☐ ☐ ☐ Very difficult

2. How helpful was MindMate in addressing your mental health concerns?

1 2 3 4 5

Extremely helpful ☐ ☐ ☐ ☐ ☐ Not helpful at all

3. How supported did you feel after using MindMate?

1 2 3 4 5

Very supported ☐ ☐ ☐ ☐ ☐ Not supported at all

4. How likely are you to use the chatbot again in the future?

1 2 3 4 5

Extremely likely ☐ ☐ ☐ ☐ ☐ Extremely unlikely

5. Did you experience any difficulties or discomfort while using MindMate?

1 2 3 4 5

No, none at all ☐ ☐ ☐ ☐ ☐ The experience was very uncomfortable

Figure 6:

6.How satisfied are you with the UI

1 2 3 4 5 6

Very Satisfied ☐ ☐ ☐ ☐ ☐ ☐ Not Satisfied at all

7.Did the music help you feel more relaxed and calm?

1 2 3 4 5

Very Calm ☐ ☐ ☐ ☐ ☐ Not Calm at all

8.How seamless was the transition to a call when using the call feature?

1 2 3 4 5

Very Seamless ☐ ☐ ☐ ☐ ☐ Not Seamless at all

Do you have additional feedback?

Your answer

Submit Clear form

Never submit passwords through Google Forms.

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Google Forms

Figure 7: Survey Images

10.1 User feedbacks

10.1.1 INTRODUCTION TO FEEDBACK FORM ANALYSIS

The feedback form was designed to gather user insights on the functionality, effectiveness, and overall user experience of MindMate. The results are instrumental in evaluating the chatbot's success and identifying areas for improvement. The feedback was collected from users, using a scale from 1 to 5, where 1 indicates a positive response and 5 a negative one.

10.1.2 EASE OF INTERACTION

Question: *How easy was it to interact with MindMate?*

Results: 1- 54.5%, 2-36.4%, 3-9.1%

The majority of users (over 90%) found it easy to interact with the chatbot, reflecting a well-designed user interface and intuitive conversational flow. The chatbot's natural language processing and empathetic design likely contributed to this positive response. However, the 9.1% who rated a 3 suggest that some areas may still cause friction, which could be refined to enhance usability further.

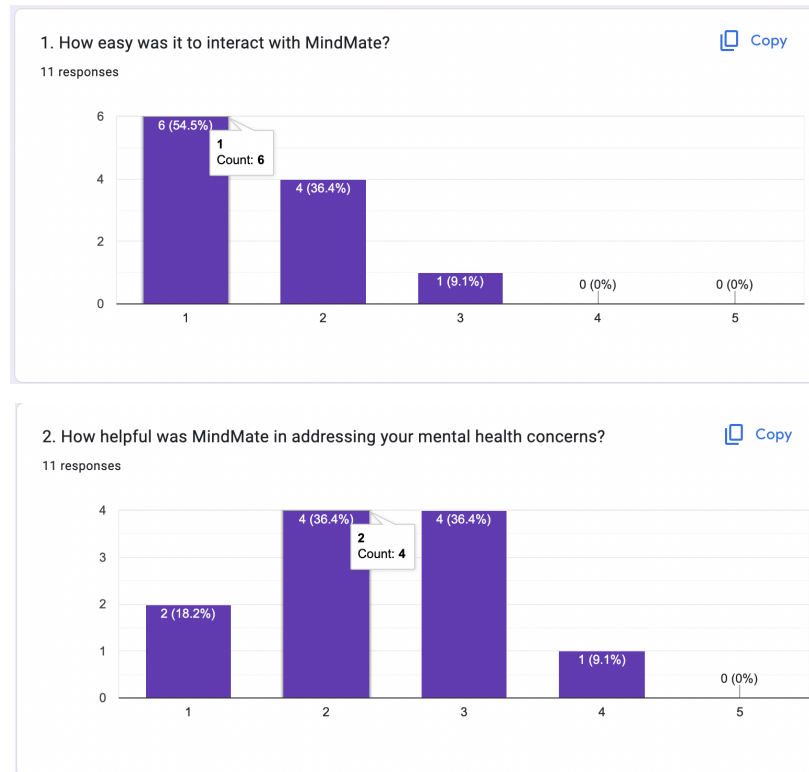


Figure 8: User feedbacks (1-2)

10.1.3 EFFECTIVENESS IN ADDRESSING MENTAL HEALTH CONCERNS

Question: *How helpful was MindMate in addressing your mental health concerns?*

Results: 1-18.2%, 2-36.4%, 3-36.4%, 4-9.1%

Over 54.6% of respondents found the chatbot helpful (ratings 1 and 2), indicating that it generally addressed mental health concerns well. However, the significant portion of ratings at 3 and 4 (45.5%) suggests that while MindMate was empathetic, it has limitations in offering comprehensive advice. This reflects the known limitations of AI in mental health, emphasizing the need for MindMate as a supplementary tool rather than a replacement for professional therapy.

10.1.4 FEELING OF SUPPORT

Question: *How supported did you feel after using MindMate?*

Results: 1-18.2%, 2-36.4%, 3-45.5%

Most users felt moderately supported, with 54.6% rating 1 or 2. However, the 45.5% who rated it a 3 indicate that there is room for improvement in making users feel more supported. Enhancing MindMate’s supportiveness could involve more dynamic follow-up questions and personalized recommendations tailored to the user’s specific input.

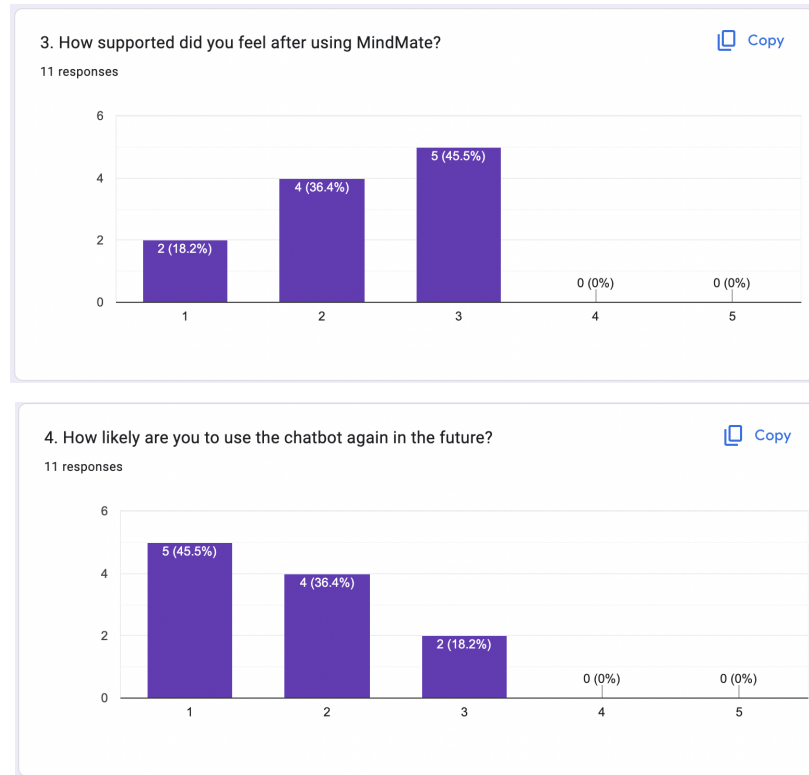


Figure 9: User feedbacks (3-4)

10.1.5 LIKELIHOOD OF FUTURE USE

Question: *How likely are you to use the chatbot again in the future?*

Results: 1-45.5%, 2-36.4%, 3-18.2%

A majority of users expressed a high likelihood of returning to MindMate, with 81.9% giving a rating of 1 or 2. This positive trend suggests that users appreciate the chatbot’s functionality and effectiveness. The neutral feedback (18.2%) might indicate areas where MindMate could further engage users or enhance certain features to increase retention.

10.1.6 USER DIFFICULTIES AND DISCOMFORT

Question: *Did you experience any difficulties or discomfort while using MindMate?*

Results: 1-27.3%, 2-45.5%, 3-18.2%, 4-9.1%

Most users reported minimal discomfort, with 72.8% giving a favorable rating (1 or 2). The 27.3% who rated 3 or higher could indicate specific issues like technical glitches or

minor interaction difficulties. Identifying these friction points can help ensure a smoother experience for future users.

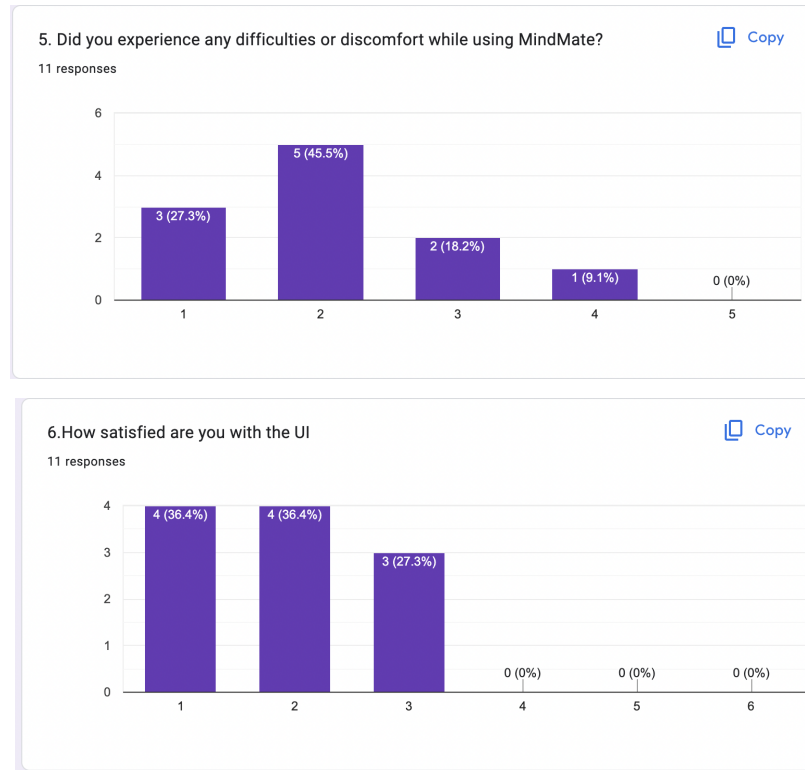


Figure 10: User feedbacks (5-6)

10.1.7 USER INTERFACE SATISFACTION

Question: *How satisfied are you with the UI?*

Results: 1-36.4%, 2-36.4%, 3-27.3%

The UI received positive feedback from the majority of users (72.8%), with bright colors and simple design choices contributing to this outcome. The 27.3% neutral rating suggests that while the interface is good, it could benefit from further refinement, such as adding more customization options like a dark mode or improving navigation features.

10.1.8 EFFECT OF BACKGROUND MUSIC

Question: *Did the music help you feel more relaxed and calm?*

Results: 1-45.5%, 2-36.4%, 3-9.1%, 4-9.1%

A significant portion of users (81.9%) felt that the background music contributed positively to their experience, helping them feel relaxed. Given its importance in creating a calming atmosphere, this feature seems to enhance the user experience. However, the few respondents who did not find the music helpful (18.2%) suggest that providing more music options or the ability to turn off the music entirely is important.

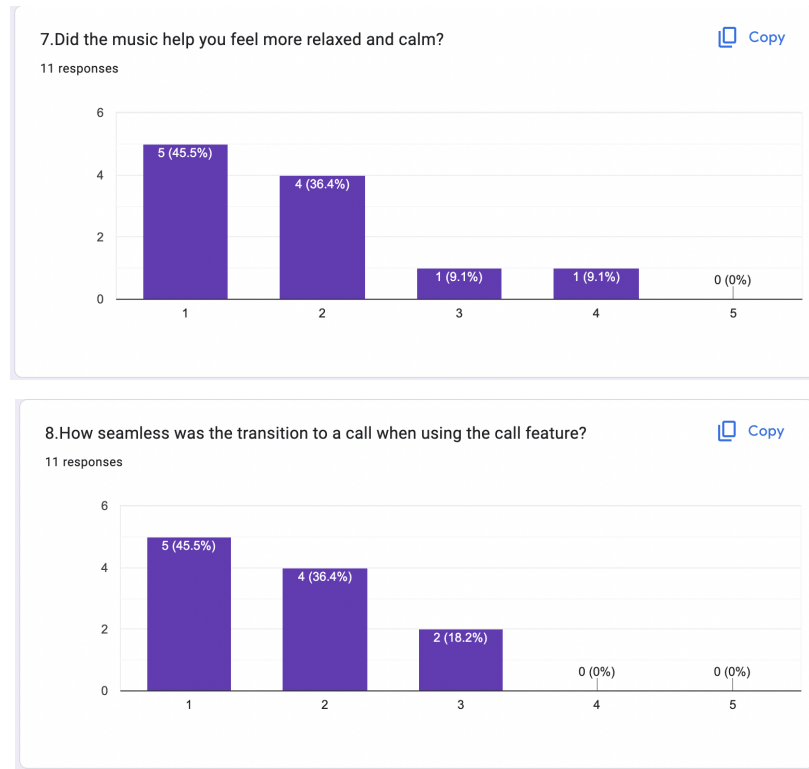


Figure 11: User feedbacks (7-8)

10.1.9 SEAMLESSNESS OF CALL TRANSITION

Question: *How seamless was the transition to a call when using the call feature?*

Results: 1-45.5%, 2-36.4%, 3-18.2%

Most users (81.9%) found the transition to the call feature seamless, which speaks to the efficiency of the system's design. The room assignment feature and email notifications allowed for smooth transitions between chatbot conversations and live calls. Improvements can be made for the 18.2% who found the process less seamless, potentially through displaying a countdown timer or providing more transparent feedback during the transition.

10.1.10 ADDITIONAL FEEDBACK (QUALITATIVE INSIGHTS)

In addition to quantitative feedback, qualitative responses provided further insights into the user experience. Common themes included the appreciation of MindMate's empathetic nature, the aesthetic appeal of the user interface, and the usefulness of the call feature. Users also made suggestions for future improvements, such as offering different music choices and enhancing the chatbot's conversational depth.

10.1.11 CONCLUSIONS

The feedback form results demonstrate that MindMate successfully provides a supportive, user-friendly, and calming platform for mental health conversations. The chatbot was well-

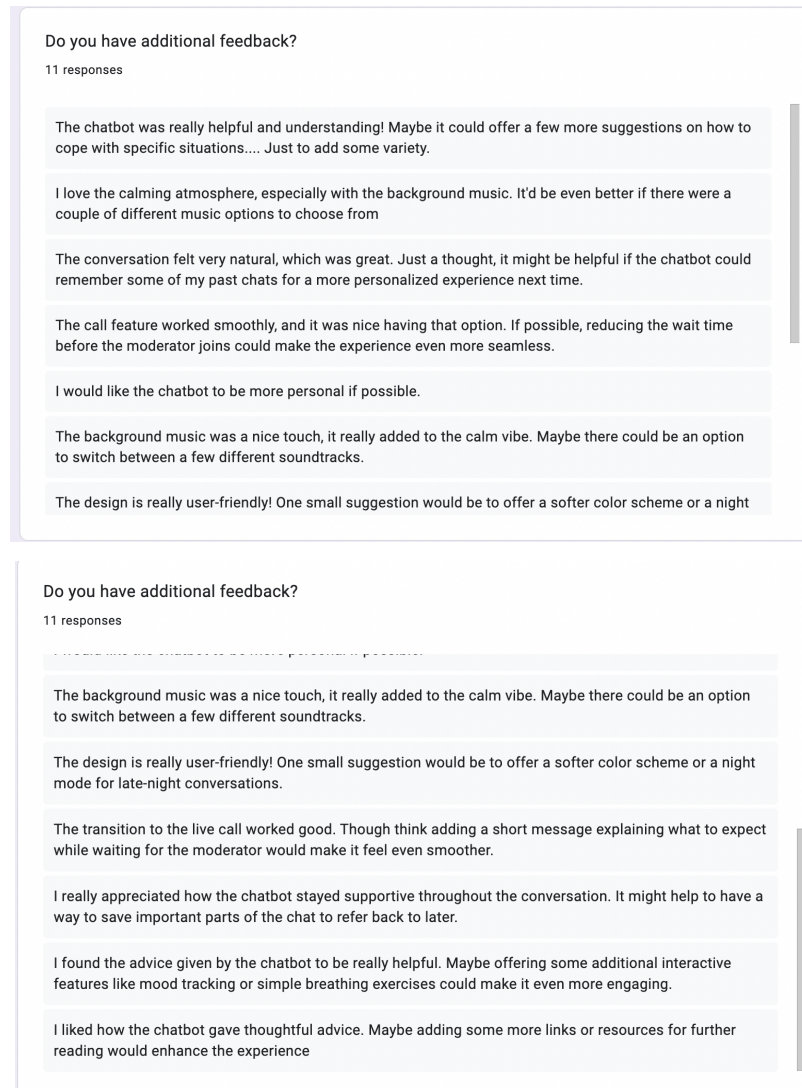


Figure 12: User feedbacks (9-10)

received overall, particularly regarding ease of use and seamless integration of features like background music and the call option. However, there are areas where improvements can be made to better address mental health concerns, enhance user support, and further optimize the user interface.

10.2 Additional User Comments

- Users expressed strong appreciation for the chatbot's helpfulness and understanding.
- Interactions were noted to feel natural and supportive, fostering a sense of empathy and connection.
- The chatbot effectively addressed users' mental health concerns.

10.2.1 ATMOSPHERE AND MUSIC

- **Calming Environment:** Users highlighted the calming atmosphere created by background music.
- **Music Options:** Many respondents suggested adding multiple music options to enhance the user experience. Customizable music choices would cater to individual preferences and moods.

10.2.2 PERSONALIZATION

- **Memory Functionality:** Users expressed a desire for the chatbot to remember past interactions to provide a more tailored experience.
- **Contextual Relevance:** Incorporating memory would allow the chatbot to offer advice based on previous discussions, enhancing the continuity of support.

10.2.3 CALL FEATURE FUNCTIONALITY

- **Seamless Transition:** Users appreciated the smooth transition to a live call with a moderator.
- **Reduced Wait Time:** Suggestions were made to reduce the wait time before connecting to a moderator for more immediate support.
- **Expectations Message:** Users recommended providing a brief message outlining what to expect during the waiting period to alleviate anxiety.

10.2.4 USER INTERFACE DESIGN

- **Positive Remarks:** The app's user interface was praised for its design and user-friendliness.
- **Color Scheme Options:** Suggestions included offering a softer color scheme or a night mode for late-night use, enhancing comfort during vulnerable times.

10.2.5 INTERACTIVE FEATURES AND RESOURCES

- **Enhanced Engagement:** Users expressed interest in additional interactive features, such as:
 - Mood tracking capabilities.
 - Simple breathing exercises.
 - The option to save important parts of the chat for future reference.
- **Further Reading:** Requests were made for links or resources for further reading to complement the chatbot's advice.
- **Empowerment Tools:** Implementing these features could empower users to manage their mental health actively and create a more engaging experience.

11 Evaluation Survey

To assess the performance and user experience of the models implemented in MindMate, an evaluation survey was conducted. The survey was designed to gather both quantitative and qualitative feedback from users, focusing on various aspects of the models such as relevance, empathy, helpfulness, and clarity. Participants were asked to compare three models—GPT 4o, LLaMA 3, and Gemini—across different dimensions and provide ratings that reflect their overall experience.

The questionnaire contained several questions where users could rate each model, as well as an additional comments section for more detailed insights. The survey helped us understand which model was the most effective in addressing user queries and concerns related to mental health, as well as their general satisfaction with the system.

11.1 Evaluation Feedback

The evaluation feedback was collected through a set of standardized questions, asking participants to evaluate the performance of each model. The following figures display the blank questionnaire that participants filled out during the evaluation process.

The questionnaire aimed to measure key factors such as:

Figure 13 displays two evaluation forms, (a) and (b), used for assessing the performance of three models: GPT-4, LLaMA 3, and Gemini.

(a) First Evaluation Form:

- 1. Relevance:** Which model best addresses your query or concern?
☐ GPT-4
☐ LLaMA 3
☐ Gemini
- 2. Empathy:** Which model demonstrates the best understanding of the user's feelings and emotions?
☐ GPT-4
☐ LLaMA 3
☐ Gemini
- 3. Helpfulness:** Which model provides the most useful information or suggestions to assist the user?
☐ GPT-4
☐ LLaMA 3
☐ Gemini

(b) Second Evaluation Form:

- 4. Clarity:** Which model delivers a response that is the clearest and easiest to understand?
☐ GPT-4
☐ LLaMA 3
☐ Gemini
- 5. Overall Quality:** Based on the user query and responses, which model do you believe provided the best overall response?
☐ GPT-4
☐ LLaMA 3
☐ Gemini
- Additional Comments:**
Your answer

Figure 13: Evaluation Forms

- **Relevance:** Which model best addressed the user's query or concern?
- **Empathy:** Which model demonstrated the highest level of understanding and emotional intelligence?
- **Helpfulness:** Which model provided the most useful and actionable suggestions?
- **Clarity:** Which model provided the clearest, most understandable responses?
- **Overall Quality:** An overall assessment of each model's performance based on the user's experience.

The collected feedback offers insights into which models were most effective for different aspects of the interaction, allowing us to make informed decisions about improving the chatbot experience.

11.2 Evaluation Results

The results of the evaluation survey were collected and analyzed to determine the performance of each model—GPT 4o, LLaMA 3, and Gemini—across various metrics. The survey responses were compiled, and pie charts were generated to visualize the percentage of users who selected each model for each question. Below is a detailed analysis of the results.



Figure 14: Evaluation Results

- **Relevance:** When asked which model best addressed their query or concern, the majority of users (80%) selected GPT 4o, highlighting its strong ability to address specific mental health concerns. LLaMA 3 was selected by 20% of users, while Gemini did not receive any selections for this category, indicating that it was less effective in addressing user queries compared to the other models.

- **Empathy:** For empathy, GPT 4o once again led with 80% of users finding it to be the most emotionally intelligent. However, both LLaMA 3 and Gemini managed to gain some recognition in this category, with 10% of users each selecting them. This suggests that while GPT 4o excels in understanding emotions, the other models still show some potential in this area.



Figure 15: results of GPT 4o

- **Helpfulness:** When it came to providing useful information or suggestions, GPT 4o was selected by 70% of users, further reinforcing its status as the most helpful model in this context. LLaMA 3 was chosen by 20% of respondents, while 10% felt that Gemini provided the most useful suggestions.
- **Clarity:** In terms of clarity, GPT 4o still maintained the majority with 50%, but it faced stronger competition from the other models in this category. LLaMA 3 was chosen by 20% of users, and Gemini performed better here than in other categories, receiving 30% of user selections. This suggests that while GPT 4o offers the clearest responses overall, some users found Gemini to be quite effective in providing understandable answers.
- **Overall Quality:** Finally, for overall quality, GPT 4o was again rated the highest, with 80% of users selecting it as the best overall performer. LLaMA 3 and Gemini each received 10% of the votes, indicating that while these models showed some promise, they were generally less favored compared to GPT 4o.

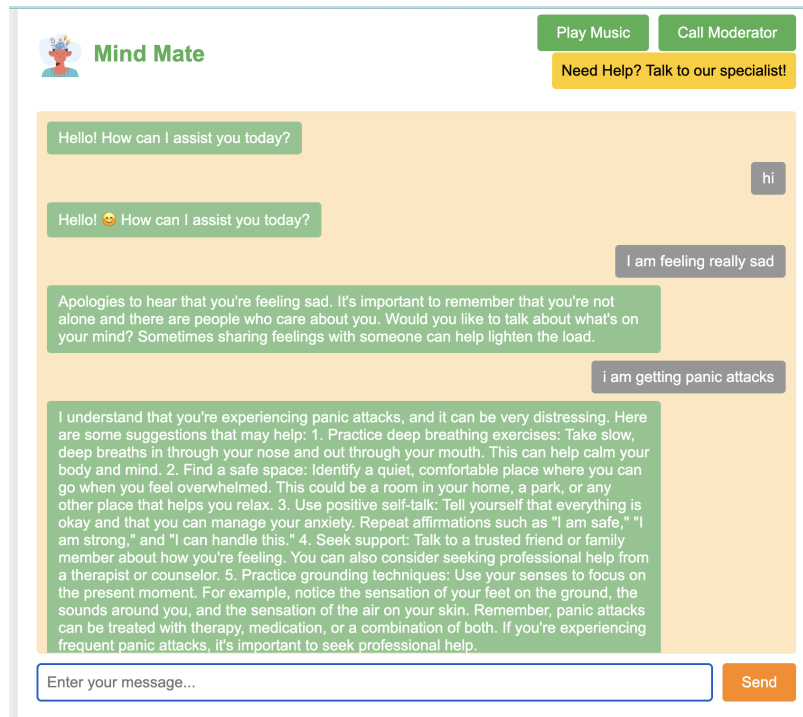


Figure 16: LLama3 result

The survey results indicate that GPT 4o is the clear leader across most categories, particularly in relevance, empathy, helpfulness, and overall quality. LLaMA 3 consistently performed as a secondary option, especially in the areas of relevance and helpfulness, while Gemini, though less favored overall, did show potential in terms of clarity. These insights will be useful in guiding future improvements to the chatbot, particularly in refining the performance of LLaMA 3 and Gemini for specific tasks.

11.3 Additional Comments

In addition to the quantitative survey results, we collected qualitative feedback from users to gain further insights into their experiences with each model. Below is an analysis of the common themes and observations from the users' comments.

- GPT 4o's Superior Performance:** A recurring theme in the user feedback was the acknowledgment of GPT 4o as the most superior model in terms of understanding context and offering meaningful, empathetic support. Multiple users commented on how GPT 4o felt more natural and human-like in conversation, particularly when addressing complex or emotional topics.

Comments such as *"GPT 4o was far superior when it came to understanding context and offering meaningful support"* and *"GPT 4o feels more natural in conversation, especially when it comes to mental health"* highlight the model's strength in providing relevant and emotionally intelligent responses. This reflects the quantitative results,

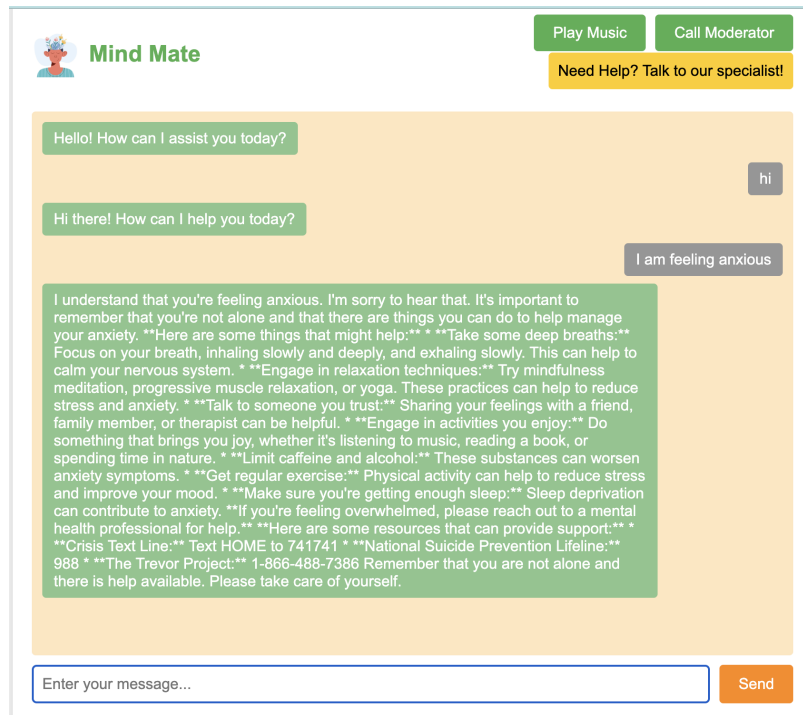


Figure 17: Gemini result

where GPT 4o consistently scored highest in categories like relevance, empathy, and overall quality.

- LLaMA 3's Strength in Coherence and Clarity:** While GPT 4o was widely praised, some users felt that LLaMA 3 performed well in delivering concise and coherent responses. Users mentioned that LLaMA 3 provided more straightforward and to-the-point answers, particularly when addressing more complex questions. One comment noted, *"LLaMA 3 provided more coherent and to-the-point answers for me, especially when I asked more complex questions"*, indicating that this model might excel in providing clear and direct information. However, a few users did mention that LLaMA 3's responses felt more robotic compared to GPT 4o.
- Gemini's Speed and Intuition:** Users offered mixed feedback regarding Gemini. Some praised its ability to deliver quick and intuitive responses, especially for shorter queries. For example, one user stated, *"Gemini did well, especially for quick responses"*, while another commented, *"I found Gemini to be the most intuitive and relatable model."* However, when it came to more in-depth or personalized interactions, users tended to favor GPT 4o. This aligns with the quantitative results, where Gemini scored lower in categories like empathy and overall quality but performed relatively better in clarity.
- General Impressions:** The overall consensus from the additional comments is that GPT 4o consistently set the standard for emotional intelligence, depth, and conver-

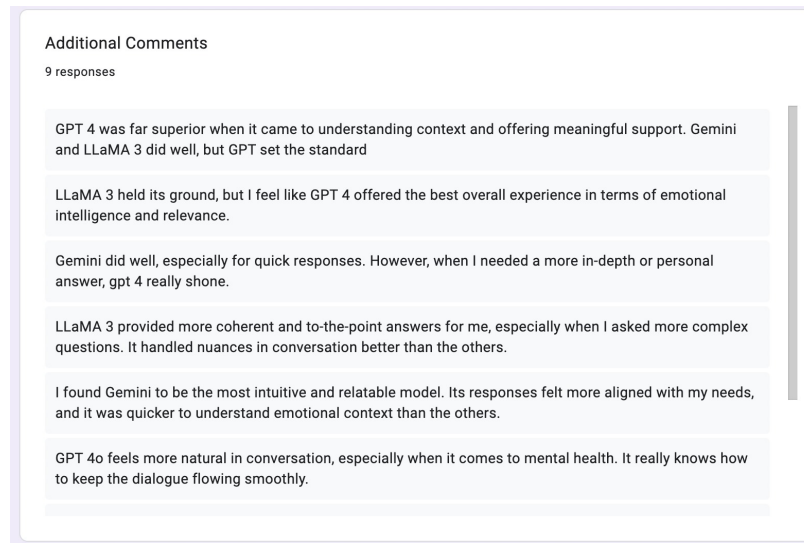


Figure 18: Additional comments in eval survey

sational flow. While LLaMA 3 and Gemini showed promise in specific areas such as clarity and speed, GPT 4o was preferred for its overall performance. As one user summarized, *"I think all three models have potential, but GPT 4 clearly outperformed."* These comments reflect the more balanced but still definitive preference for GPT 4o seen in the survey results.

In conclusion, the user feedback reinforces the quantitative findings: GPT 4o stands out as the most well-rounded and effective model for addressing mental health concerns, while LLaMA 3 and Gemini are recognized for their strengths in certain contexts, such as clarity and speed, respectively.

12 Conclusion

The development of MindMate as part of our course on "Programming with LLMs" has provided invaluable insights into the application of large language models in the domain of mental health care. Throughout this project, we have gained a deeper understanding of how LLMs can be harnessed to create meaningful interactions that simulate human-like empathy and support.

By exploring various frameworks designed to optimize the use of LLMs, such as Langchain, we have been able to refine our approach, even though we chose not to incorporate DSPy into our implementation. This decision allowed us to focus on tailoring MindMate to effectively meet the specific needs of users seeking mental health support, thereby enhancing its functional capabilities.

The course has emphasized the potential of LLMs to revolutionize traditional systems of care by offering scalable, accessible solutions. The success of MindMate demonstrates that well-designed AI applications can bridge the gap between technological advancements and

the human touch that is essential in therapeutic contexts. As we reflect on our journey, it is clear that programming with LLMs not only equips us with technical skills but also challenges us to consider the ethical implications and responsibilities associated with deploying AI in sensitive areas like mental health.

In conclusion, the knowledge and experience gained from this project will undoubtedly inform our future endeavors in the field of AI, as we continue to explore innovative applications that can enhance well-being and support individuals in their mental health journeys.

Appendix A. Statement of Contribution

A.1 Collaborative Dataset Acquisition

Our group embarked on this project with a clear plan for equitable collaboration from the very beginning. Once we identified the problem we aimed to tackle, we collectively brainstormed and developed a strategy that would ensure all voices were heard and every member contributed meaningfully to the outcome.

The first step in our project involved searching for relevant datasets that would underpin our research and model training. We worked together to identify potential sources, evaluate their relevance, and ultimately secured access to the datasets that would best suit our needs. This collaborative effort was essential as it allowed us to pool our resources and insights, leading to a more comprehensive selection of datasets.

A.2 Model Selection and Research

Following the acquisition of the datasets, we delved into research to determine the most suitable models for addressing our specific problem. Through extensive discussions and evaluations, we identified three prominent models: GPT-4, LLaMA 2, and Gemini. Each model was selected based on its unique strengths and potential effectiveness in handling the nuances of mental health conversations. To maximize our efficiency and expertise, we assigned each team member to focus on one of the three models, allowing us to dive deeper into their specific features and capabilities.

A.3 Training the Models

With our roles defined, each member dedicated substantial time and effort to train their respective model on the curated datasets. This phase involved not only the technical aspects of model training but also iterative testing and refinement to ensure optimal performance. We shared insights and strategies throughout this process, fostering an environment of continuous improvement and collaboration.

A.4 Division of Additional Tasks

In addition to model training, we recognized the need to divide other essential components of the project to maintain progress and enhance our overall productivity. We split responsibilities for developing the user interface (UI), implementing the call feature, and performing data cleaning. Each of us brought our individual strengths to these tasks, ensuring that the project components were not only functional but also user-friendly and effective.

A.5 User Feedback and Survey Creation

To gauge the performance of our chatbot, MindMate, and the underlying models, we devised a user review survey that focused on the experiences of individuals interacting with both the best-performing models and MindMate itself. We reached out to a diverse group of acquaintances, including friends, family, and colleagues, to collect a wide range of feedback. This outreach was instrumental in understanding user perceptions and identifying areas for improvement.

A.6 Reflection

Each team member engaged actively throughout the project phases, ensuring that we remained aligned and supportive of one another's efforts.

Our collaborative approach not only strengthened our final product but also fostered a sense of camaraderie and shared responsibility within the group. As we move forward, the insights gained from this project will undoubtedly influence our future endeavors in the field of artificial intelligence and mental health.

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